GRIMM AEROSOL Technik GmbH & Co. KG

Portable Dust Monitor

SERIES 1.100

Table of Content

General

Remark

Many Hard- and Software- Labels, mentioned in this manual are registered trademarks and must be understood as such.

The author tried to provide complete and correct information within this manual. However Grimm Aerosol Technik GmbH doesn't warrant completion and correctness for the given information. Grimm won't therefore be responsible for any damage resulting from applying information given in this manual.

Grimm Aerosol Technik GmbH won't be responsible for patent infringement or rights against third parties resulting from use of this document.

As a matter of fact hard- and software is continuously improved. Therefore discrepancies between this document and actual behavior of Hard- und Software might appear. In this case please request an actual document.

Take your soft- and/or hardware only into operation after studying this manual! The manufacturer and/or supplier won't be held responsible for damage resulting from improper usage.

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This document relates to:
Portable Dust Monitor SERIES 1.100
GRIMM_1100_Me_KM_260805

Symbols used within this manual

To help you find quickly important aspects the following signs are used.

This Symbol informs you about potential hazards which may even cause malfunction or damage to the hardware and advises you how to avoid this.

This Symbol stands for useful hints which might simplify and optimize your work process.

1 Advisories and Warnings

GRIMM Technologies, Inc. reserves the right to modify equipment in this document without prior notification of the user. Any improvement or technical modifications may have insignificant deviations between the descriptions in this instruction and the actual instrument provided.

Instrument contains a Class 3-B Laser!

- The manufacturer accepts no direct or indirect liability, if the operator opens or uses the instrument for any purpose other than the manufacturer's intended use. This instrument complies with European Standard EN 61010 and EN (DIN VDE 0411 T1) protection for electronic measuring instruments. The instrument outlined in this document is manufactured and tested in accordance to European Work and Safety Regulations. To ensure proper operation in the field, it is the user's responsibility to read and follow all instructions and adhere to all warnings as stated in this manual.
- The equipment is intended to operate only with the specified power supply and designated battery.
- The battery provided is to be charged only with power supply included with the particle size analyzer/dust monitor.
- The lithium battery (SL-389, 3.6V 1AH) provided for the internal clock, is not intended to be recharged and must be changed by an authorized service technician.
- The user must not open the power supply. In the event of a malfunction or an error reading, please contact GRIMM for repair.
- When safe operation of the unit is no longer possible, the user must power down the unit immediately.
- The instrument must only be opened by factory authorized service personnel as opening the laser unit can cause the risk of exposure to Class 3-B laser-radiation (See EN 60825 DIN VDE 0837 T1).
- The glass fuses built in the instrument (Pico fuse 2A, quick-type, and switch-fortunes 300A /32V DC) should be replaced only by authorized service personnel.

After transportation or storage of the instrument at low temperatures, allow the instrument sufficient time to warm up prior to operation. Failure to do so can cause condensation in the units pumping mechanism which will cause the pump to malfunction. In this event, the unit will automatically shut down.

Never operate the unit without PTFE filter because the PTFE filter is before pump for protection!

1.1 Battery maintenance

The following recommendations are made to ensure optimal operation and lifespan of the battery:

Never shorten the two contacts of the battery.

- Discharged batteries should never be charged during the normal operation of the dust monitor.
- Operate the particle size analyzer only with the power supply provided.
- Do not over charge the battery.
- Completely discharge the battery before recharging.
- Charge the battery as soon as possible after a complete discharge.
- Store the battery in a completely charge state when possible.
- Batteries should be recycled when their life span is complete.
- Remove the battery from the unit when using alternative power sources.

The battery supplied contains toxic heavy metals and must be disposed in accordance with all state, local and federal regulations.

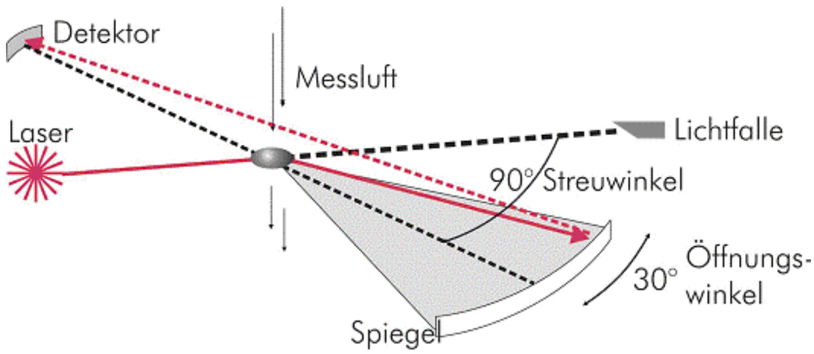
2 General Principles of Operation

The dust monitors of the Series 1.100 are small portable units, used for the continuous measurement of particles in the air (aerosols). These particles can be reported in various modes. However, these measurements are determined in one of two basic modes, **particle concentration** as counts/l or **mass concentration** as µg/m^3. These measurements are reported for the various size distribution channels.

All units of the series 1.100 use a light-scattering technology for single-particle counts, whereby a semiconductor-laser serves as the light-source. The scattered signal from the particle passing through the laser beam and is collected at approximately 90° by a mirror and transferred to a recipient-diode. The signal of the diode passes, after a corresponding reinforcement, a multi-channel size classifier. A pulse height analyzer then classifies the signal transmitted in each channel. These counts can be displayed and are also stored in the data storage card and may be transferred via the RS 232 for further analysis.

The ambient-air, to be analyzed, is drawn into the unit via an internal volume-controlled pump at a rate of 1.2 liters/minute. The sample passes through the sample cell, past the laser diode detector and is collected onto a 47-mm PTFE filter. The entire sample is collected on the PTFE filter, which can then be analyzed gravimetrically for verification of the reported aerosol's mass. Additionally, further chemical analysis can be performed on the deposited residue. The pump also generates the necessary clean sheath air, which is filtered and passes through the sheath air regulator back in to optical chamber. This is to ensure that no dust contamination comes in contact with the laser-optic assembly. This particle free airflow is also used for the reference-zero test during the auto-calibration.

At the beginning of each measurement, the instrument initiates a self-test, which last approximately 30 seconds. The actual dust-measurement begins when the LCD displays the first result. Subsequent results will be displayed every 6 seconds. The real-time dust-concentration measurement data are transferred each minute to the removable data storage-card, if used. The data is also available via the built-in RS-232 serial port. This data may be transmitted to an external computer or printer. This data is available in intervals of every 6 seconds (fast-mode) or every 60 seconds (normal mode). Other options for data retrieval times are available via the PC software.



Optical measuring principle

3 Prortable GRIMM Dust Monitors Series 1.100

3.1 Common technical data for the units 1.104–1.109

measuring principle:	90° light-scattering
classification:	in 8 , 15 or 31 channels (1.105, 1.108, 107 or 1.109). The indicated channel thresholds apply to a counting efficiency of 50% with monodispers latex aerosol.
reproducibility:	±2% over the whole measuring range.
self test:	Automatically after every start.
sample flow rate:	1.2 l/min ±5% constant with controller.
rinsing air rate:	0.3 l/min constant with controller.
Sample air filter:	47 mm PTFE - round filter (without supporting tissue).
operation:	over transparency keyboard or over PC and RS-232-interface.
LCD:	2 x 16 alphanumeric signs, illuminated
data output LCD:	Dates as sliding average over one minute or average with associated sample volume. Alarm values, Battery capacity, gravimetric factor, location number, date and time, dates of the optional ancillary equipment (sensors).
analog input:	3 pieces (0-10V), resolution 10 Bit (approx. 10 mV).
data interface:	ASCII: RS-232 (9600 baud, 8 bit, no par parity, 1 stop-bit, protocol: Xon/Xoff)
data storage:	-intern 80 kByte -PCMCIA-Card from 256 kByte up to 6 MByte, battery backed.
power supply:	-battery: 12V/2,3Ah, Typ LC-SA122R3B6 Panasonic, full charged for 7...8 hours continuous operation. -battery: LCTA 121R9PG, 12/1,9Ah, Panasonic. -18V-power supply: MI2818, AC 95-250V, 47-63Hz.

operation temperature range:	0...+40 °C, r.H. < 95 % (no condensation).
transporting temperature range:	-20...+50 °C, r.H. < 95 % (no condensation).
sample air:	+4...+40 °C, r.H. < 95 % , **no corrosive or explosive gas portions.**
pressure range of the sample air:	0...-50 mbar (short-time). Over pressure (max.+100 mbar) and on longer measurements with high under pressure (max. -100 mbar) use the sample air feedback.
housing:	
dimensions:	24 x 12 x 6 [cm]
weight:	1.7 kg + battery (0.7 kg).

Different model types of the units 1.104 – 1.109

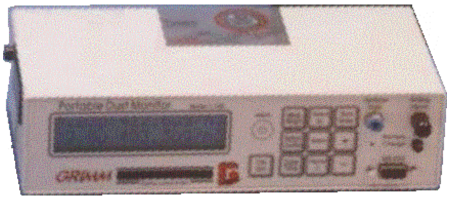
Reading on the LCD

<u>1.101</u>

Inhalable	inhalable dust, up to the highest measurable cannel (by extrapolation the measuring range is between 0.4-15.0 µm)
Thoracic	dust fraction, over the larynx up to the bronchi.
Alveolic	respirable dust fraction, reach the alveoles (finest fraction)
PM_{10}	Fraction to US-EPA
$PM_{2,5}$	Fraction to US-EPA
Particle	**particle concentration** in 16 measuring channels applied to the volume of one litre sample air.

<u>1.104</u>

Inhalable	inhalable dust, up to the highest measurable cannel (by extrapolation the measuring range is between 0.4-15.0 µm)
Thoracic	dust fraction, over the larynx up to the bronchi.
Alveolic	respirable dust fraction, reach the alveoles (finest fraction)
PM_{10}	Fraction to US-EPA
$PM_{2,5}$	Fraction to US-EPA



1.105

Inhalable	inhalable dust, up to the highest measurable cannel (by extrapolation the measuring range is between 0.4-15.0 µm)
Thoracic	dust fraction, over the larynx up to the bronchi.
Alveolic	respirable dust fraction, reach the alveoles (finest fraction)
PM_{10}	Fraction to US-EPA
mass	**mass concentration** with the channel limits 1-2-5-10-15µm greater than (>) or smaller than (<) the limit (by extrapolation the measuring range is between 0.5-18.5 µm
particle	**particle concentration** in 8 measuring channels applied to the volume of one litre sample air.

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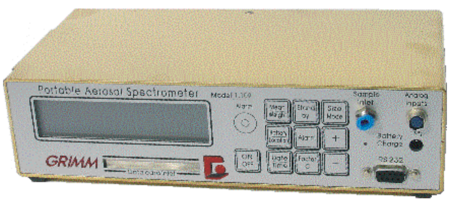
PM_{10}	Fraction to US-EPA
$PM_{2,5}$	Fraction to US-EPA
PM_1	Fraction to US-EPA
mass	**mass concentration** with 31 size channels from 0.25 - >32 µm in µg/m³
particle	**particle concentration** in 31 measuring channels applied to the volume of one litre sample air.

1.108

mass	**mass concentration** with 15 size channels from 0.3 - >20 µm in µg/m³
particle	**particle concentration** in 15 measuring channels applied to the volume of one litre sample air.



1.109

mass	**mass concentration** with 31 size channels from 0.25 - >32 µm in µg/m³
particle	**particle concentration** in 31 measuring channels applied to the volume of one litre sample air.

<table>
<tr><td rowspan="3">
</td><td colspan="2"><u>7.309 [FINE]</u></td></tr>
<tr><td>mass</td><td>mass concentration with 12 size channels from 0.27 - >2.0 µm in µg/m³</td></tr>
<tr><td>particle</td><td>particle concentration in 12 measuring channels applied to the volume of one litre sample air.</td></tr>
<tr><td rowspan="3">
</td><td colspan="2"><u>7.310 [COARSE]</u></td></tr>
<tr><td>mass</td><td>mass concentration with 12 size channels from 0.7 - >50 µm in µg/m³</td></tr>
<tr><td>particle</td><td>particle concentration in 12 measuring channels applied to the volume of one litre sample air.</td></tr>
</table>

3.2 Model overview

Model	measuring range		Size channels [P/l = Particle/Litre]	applications
	[µm]	[µg/m^3]		
1.101 „Junior“	0,5...>15 sample air: 0,6 l/min	1...>10.000	-EN 481: (inhalable, thoracic, respirable) US-EPA:(PM-10,PM-2,5)	Occupational health and safety, environmental measurements
1.104 „Workcheck-4”	0,5...>15 sample air: 1,2 l/min	1...65.000	-EN 481: (inhalable, thoracic, respirable) US-EPA:(PM-10,PM-2,5)	Occupational health and safety, environmental measurements
1.105 „Workcheck-5“	0,5...>15 sample air: 1,2 l/min	1...65.000 particle counts: up to 500.000 P/l	-EN 481: (inhalable, thoracic, respirable) US-EPA:(PM-10) -8 channels µg/m3 for µm: >0,5/1/2/3,5/5/7,5/10/15 -8 channels in P/l* for µm: >0,75/1/2/3,5/5/7,5/10/15	Occupational health and safety, environmental measurements filterchecks
107 „Environment“	0,25...>32 sample air : 1,2 l/min	1...100.000 particle counts: up to 2.000.000 P/l	-31 channels in P/l* for µm or in µg/m³: (0,25/0,28/0,3/0,35/0,4/0,45/ 0,5/0,58/0,65/0,7/0,8/1/1,3/ 1,6/2/2,5/3/3,5/4/5/6,5/7,5/8,5/10/12,5/ 15/17,5/20/25/30/32) US-EPA:(PM-10,PM-2,5, PM-1)	environmental measurements
1.108 „Aircheck“	0,3...>20 sample air : 1,2 l/min	1...>100.000 particle counts: up to 2.000.000 P/l	-15 channels in µg/m^3 in µm: 0,3/0,4/0,5/0,65/0,8/ 1/1,6/2/3/4/5/7,5/10/15/20 -15 channels in P/l: same size in µm like above, -with the software accordant EN481 and EPA	Occupational health and safety, environmental measurements filter checks, air quality check
1.109	0,25…>32 sample air : 1,2 l/min	1…100.000 particle counts: up to 2.000.000 P/l	-31 channels in P/l* for µm or in µg/m³: (0,25/0,28/0,3/0,35/0,4/0,45/ 0,5/0,58/0,65/0,7/0,8/1/1,3/ 1,6/2/2,5/3/3,5/4/5/6,5/7,5/8,5/10/12,5/ 15/17,5/20/25/30/ 32)	Paricle counting on the workplace, Filter check, air quality check
7.309 „Filtercheck“ -FINE-	0,27...>2,0 sample air : 2,0 l/min	particle counts: up to 2.000.000 P/l	-12 Kanäle in P/l in µm: 0,27/0,3/0,35/0,4/0,45/0,5/ 0,6/0,7/0,8/0,9/1,0/1,6/>2,0	filtertests to ASHRAE
7.310 „Filtercheck“ -COARSE-	0,7...>50,0 sample air : 1,2 l/min	1...>150.000	-12 Kanäle in µg/m^3 und in P/l in µm: 0,7/ 1/ 2/ 3,5/ 5/ 7,5/ 10/ 15/ 20/ 30/ 40/ 50	Filtertests for coarse partcles e.g. car filter ISO 5011

P/l= particle counting in particle/litre*
[1]= the weather housing can be used for all units 1.104 up to 1.108.
1.164 weather housing: just with installed heater (winter operation) and ventilation (summer operation)
1.165 weather housing like 1.164 but extra with installed air dryer for the measuring operation also by high humidity (fog).

<u>Another measuring systems:</u>

1.400 stationary dust indicator:

measures airborne particles as TSP (Total Suspended particles) in the range: 0,15...1.500 µg/m^3 or 0,15...15.000 µg/m^3 with analog outputs and adjustable alarm value relies.

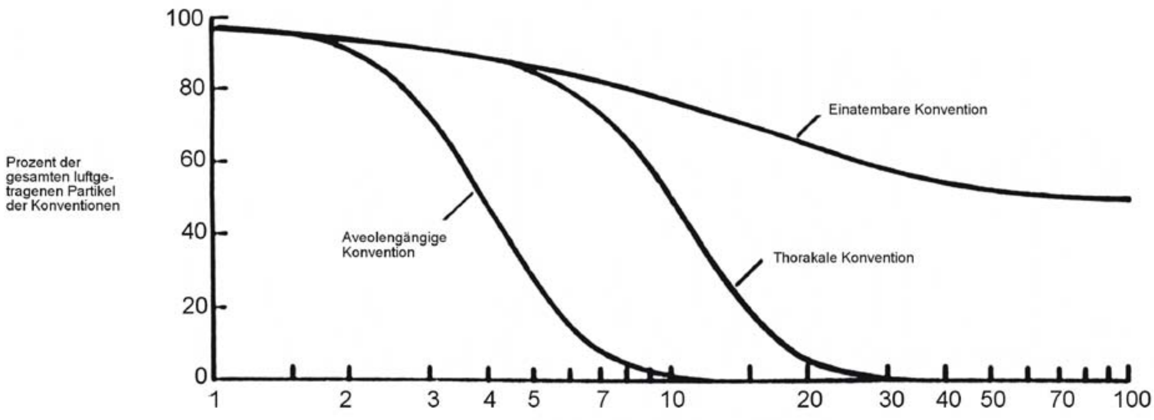
5.400 ultrafine partcle counter:

measuring system for airborne partcles in the range of 5...1.000 nm

3.3 Numeric values of the conventions as percentage of all airborne particles

Aerodynamic diameter in µm	inhalable (%)	Thoracic (%)	respirable (%)	PM-10 (%)	PM-2,5 (%)
0,0	100	100	100	100	100
1,0	97,1	97,1	97,1	100	99,5
2,0	94,3	94,3	91,4	94,2	85,5
2,5					48,0
3,0	91,7	91,7	73,9	92,2	6,7
4,0	89,3	89,0	50,0	89,3	0
5,0	87,0	85,4	30,0	85,7	-
6,0	84,9	80,5	16,8	81,2	-
7,0	82,9	74,2	9,0	75,9	-
8,0	80,9	66,6	4,8	69,7	-
9,0	79,1	58,3	2,5	62,8	-
10,0	77,4	50,0	1,3	55,1	-
11,0	75,8	42,1	0,7	46,5	-
12,0	74,3	34,9	0,4	37,1	-
13,0	72,9	28,6	0,2	26,9	-
14,0	71,6	23,2	0,2	15,9	-
15,0	70,3	18,7	0,1	4,1	-
16,0	69,1	15,0	0	0	-
18,0	67,0	9,5	-	-	-
20,0	65,1	5,9	-	-	-
25,0	61,2	1,8	-	-	-
30,0	58,3	0,6	-	-	-

Graph:



Size distribution curve of the dust particles according to ***EN 481Dust on the workplace***

Basic data:

TSP	100 % for all particle sizes
PM-10	100 % for all particle smaller than 1,0 µm 0 % for all paricle higher than 15 µm
PM-2,5	100 % for all particle smaller than 0,5 µm 0 % for all paricle higher than 3,5 µm
PM-1 **(Entwurf)**	100 % for all particle smaller than 0,1 µm 0 % for all paricle higher than 2,5 µm

Laser-Data of the units 1.104 – 1.109

Laser type: **semiconductor - laser diode**
Laser class: **3B**

Model	**wavelength**	**P_{max}**	**$P_{nom.}$**
1.101	650 nm	5 mW	1,2 mW (key down)
1.104	780 nm	5 mW	3 mW (key down)
1.105	780 nm	5 mW	3 mW (key down)
107	683 nm	50 mW	0,5/30 mW CW (multiplex)
1.108	780 nm	40 mW	0,75/30 mW CW (multiplex)
1.109	683 nm	40 mW	0, 5/30 mW CW (multiplex)
7.309	683 nm	50 mW	0,5/30 mW CW (multiplex)
7.310	780 nm	5 mW	0,75/30 mW CW (multiplex)

3.4 Analyze software 1.177

GRIMM's 1.177 software is fully compatible with Windows™ 95/98/ME/NT 4.0/2000/XP. This program works bi-directional, this means all actions can be activated and controlled via this software. The data are in numerical or graphical presented in the following formats

Environmental Mode:	PM_{10}, $PM_{2.5}$ and PM_1 conventions.
Occupational Mode:	Inhalable, Thoracic and Respirable (alveolic).
Mass mode:	Microgram/m^3
Particle counts:	particles/liter
External Sensors	Depending on type
Service data:	Pump, battery, operational errors, etc..

GRIMM's 1.177 Software also offers options for complete statistical analysis of data as well as instrument performance and complete system diagnostics. A very detailed Software manual comes with the program.

3.5 List of accessories

Required Accessories

1.1XX	**Portable Real Time Dust Monitor**
1.110	Lead Storage Battery Type LCS 2312 AVBNC, 12V/2,3A for 7-8 h by continuous operation
1.111	Radial symmetric sampling head for TSP by 1.12 m/s
1.112A	External power supply 95-250 VAC, 47-63 Hz
1.113A	Package - 47 mm PTFE Filters – 25 pieces
1.118A	Operation Manual in English or German
1.119	Air inlet connector
1.143[E o. F]	RS-232 → Serial port connecting cable, dust monitor to computer
1.142.2	Data Storage Card with 1024 kByte storage capacity

Recommended accessories

1.144	Carrying case, Pelican ™ hard shell
1.145	Shoulder strap
1.146	Cyclone for 1.104 and 1.105; from 10 µm
1.147	Extern visual and acoustic alarm
1.148	Mini-filter for 0-test
1.149A	Big spare part set (1.110, 1.113A, 1.117)
1.151A	Isokinetic sampling head with 4 different colored probes, measuring range from 0,5-8 m/s, with tripod and 1 meter tube.
1.152	Isokinetic sampling pipe with 4 different colored changeable stainless steel probes for 2 - 25 m/s (for 1.104, 1.105 , 1.108 and 1.109)
1.153	Sensor for humidity (0 –100 %) and temperature (0,3 - +80 °C) with cabel and connector
1.154	Like 1.153 but with air velocity sensor (0,3 bis 20 m/s) too
1.162	Plug for the analog input
1.165	Weather housing with dryer and mixer electronic
1.165FG	Field Housing (fiberglass) complete with temperature control, ventilation system and moisture compensation controlling unit, 1.153 Temperature and Humidity sensor

Source of supply

Accessories and expendable items are available by

Fa. GRIMM Aerosol Technik GmbH & Co. KG
Dorfstrasse 9
83404 Ainring
Tel.: +49 (0) 8654-578-0

or by the responsible salesman.

4 Operational Overview

Please READ the manual first, than start up the dust monitor!

4.1 LCD

The optical display on the unit consists of a Liquid Crystal Display (LCD) containing two rows with 16 characters in each row. The LCD is capable of displaying the following information:

- Real time measurements for 2 channels
- Date and Time
- User Selected Location as a two digit number
- Current Battery Status as % charged
- C-Factor
- Filter Status
- Analog Inputs (Temperature and Humidity sensors are typical options)
- Outputs via the function Keys
- Error Messages

4.2 ON/OFF Key

The ON/OFF key is used to power up and power down the instrument. This key has a fail-safe mechanism to assist in preventing the instrument from being inadvertently turned off. In order to power down the instrument, the ON/OFF key must be held down for approx. one second.

To power up or power down the instrument, hold the ON/OFF key down until you hear a “beep". A re-start should not be attempted for at least five seconds after the instrument has been in the „OFF“ mode.

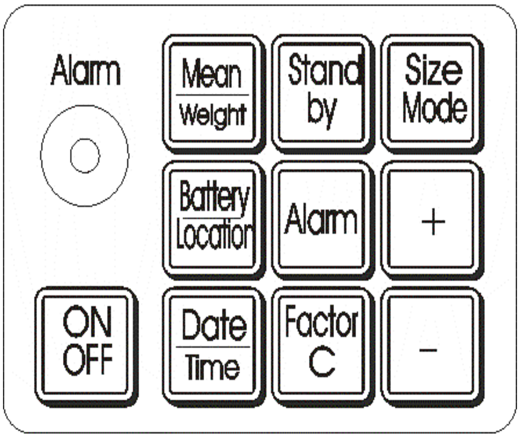
The instrument should always be placed in the stand-by mode prior to powering down the unit. Failure to do so could cause the loss of data from the memory card. In the event that the instrument is switched off due to a power failure, the unit will automatically restart measurements with the user-selected parameters once power is restored. The instrument will switch directly to the RUN mode without asking for "filter-exchange" information and will calculate the mean values included in the previous measurement.

Never turn the instrument off while in RUN mode!

4.3 Memory-functions

The function once selected in the last STANDBY mode will automatically boot up again once the instrument is removed from the STANDBY mode or powered back up. The display mode remains as before. Alarm-values, calculated filter-weight and sample flow remain stored. Should a power failure occur during a measurement cycle, all of the functions pre-selected by the user and any of the recorded values will be stored. Once power is restored, pre-selected measurement cycle will automatically be continued.

4.4 Function Keys

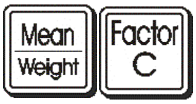


Various functions and outputs of the instrument can be viewed by pressing down the function touch keys. These keys display varying parameters depending on whether the instrument is in RUN or STANDBY mode.

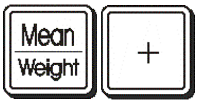
4.5 Operation of the Function Keys in the STANDBY mode

4.5.1 Mean/Weight Key

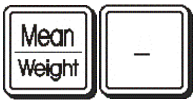
The available options for the mean/weight key are the same in the STANDBY mode as they are in the RUN mode: By pressing the with the **[Mean/Weight]** the average collected dust mass in µg/m3 will be displayed. The selected channel is hereby underlined.



When pressing **[Mean/Weight]** and the **[Factor C]** key simultaneously the message " clear memory card, are you sure? " will show up. If you are sure you have to erase, press the **[+]** key. If no memo card is in the instrument the message "clear memory, are you sure?" Will show up. The erase is the same procedure will clear the 80 Kbytes internal memory.



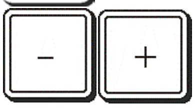
When pressing **[Mean/Weight]** and holding the **[+]** key for 5 sec the total filter weight and the sampled air will be displayed.



When pressing **[Mean/Weight]** and holding the **[-]** key for 5 sec the mean value will be reset to Zero. Also the sampled air will be reset to zero.

4.5.2 Size Mode Key

By continually pressing of the **[Size mode]** key and the **[+]** or **[-]** key, the available measurement modes will alternate on the LCD screen. When the desired measurement mode appears, simply release the **[+]** or **[-]** key.



Two channels can be selected the following way:

By continually pressing of the **[Size mode]** key and the **[+]** key, the desired measurement modes will shop up on the LCD screen. . Now release the **[Size mode]** . The message "please use + - " will show up. Now press of the **[Size mode]** key again q and press the **[+]** or **[-]** key until you have the desired 2nd channel.

All the data transmitted over the RS-232-port as well as to the data storage card are from the measurement mode as pre-selected by the user. Mass values cannot be switched to counts nor can counts be switched to mass. The desired measurement mode **must be** selected prior to the starting the measurement process.

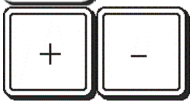
By pressing the **[Size Mode]** key alone for longer than three (3) seconds, the values of the analog-inputs from the attached sensors and the time are shown. If no sensors are attached, input 1 through input 3 will appear with no values. This mode of display remains until any other key is pressed.

4.5.3 Battery/Location Key

Pressing this key will allow the LCD to display the current battery power shown as capacity in percentage and the selected location as a number.

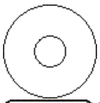
With A fully charged battery, the display can read more than 100% AND when using an external power supply the value shown will always read 130%.

By pressing the **[Battery/Location]** key and than in addition the **[+]** or **[-]** keys, the location number is changed. This number serves as identification for the locations pre-selected by the user for particulate measuring.

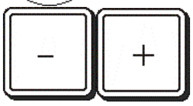


Location numbers can be selected between 1 and 99.

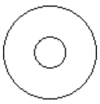
4.5.4 Alarm Key



By pressing the **[Alarm]-**key and than in addition the **[+]** or **[-]** key the alarm warning limits can be changed. Selecting a value of zero deactivates the alarm.

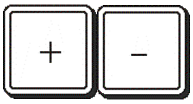


Note the value always corresponds to the pre-selected size mode for mass or particle concentration.



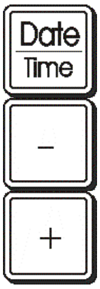
When pressing simultaneously the **[Mean/Weight]** and the **[alarm]** key the data on the data logger card will be send results to the RS-232 port for printing, etc... The instrument will show the message " I am busy... sending data". The all data send, the message "transmit data finished" will show up.

4.5.5 [+] And [-] Keys



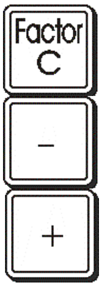
These buttons will show possible email message's

4.5.6 Date/Time Key



Depressing this key shows the date and time. Holding the key down will activate the reset date and time program. Once the reset program is activated, the date and time can be changed by pressing the **[+]** or **[-]** key. The value to be changed will be indicated by the blinking cursor. Through pressing the **[Date/Time]** key again the cursor is moved to the next digit. During any adjustment to the minute value, the value for seconds resets to zero. If there is no activity (key pressed), STANDBY mode is activated.

4.5.7 Factor C Key



Pressing this key shows the current selected gravimetric correction factor. By pressing the **[Factor C]** key and than in addition the **[+]** or **[-]** key, the gravimetric correction factor can be changed. Values can be adjusted in increments of 0.05 for values between 0.1 to 9.9. The gravimetric correction factor is preset at the factor to 1.0. Please note that the C-Factor is applied to the displayed values; all data stored on the data storage card or transferred to the RS-232 are not corrected by this value.

4.6 Operation of the Function Keys in the RUN mode

4.6.1 Mean-Weight Display Key

Pressing the **[Mean-Weight]** key in the RUN mode allows the user to view all times the mean dust-concentration as well as the total volume of air sampled during the current measurement. In the particle-concentration mode or the mass-concentration mode, the two pre-selected size channels will be displayed with their mean value for the current measurement cycle.

If the user holds down the **[Mean-Weight]** key, they will be able to view the respective mean-air volume corresponding to the mean values. These values will appear in approx. 5 seconds and remain displayed as long as the button is held down.

4.6.2 Standby Key

Pressing this key interrupts the measurement and switches the instrument to the **STANDBY** mode. Pressing this key again re-activates the measurement process.

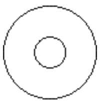
4.6.3 Size Mode Key

The size mode key is deactivated during the RUN mode.

4.6.4 Battery/Location Key

Functions the same as in the standby mode.

4.6.5 Alarm Key



Pressing this key will display the current user selected warning limits.

4.6.6 Date/Time Key

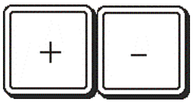
Pressing this key will display the current date and the time.

4.6.7 Factor C Key

Pressing this key will display the pre-selected gravimetric-correction factor .

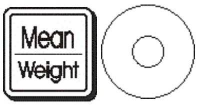
If external sensors are connected to the analog input during the RUN mode, the values for these inputs will also be displayed on a revolving cycle.

4.6.8 [+] and [-] Keys



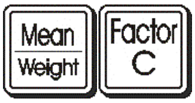
When pressing **[+]** key, the external Sensor values will be shown. The results are rolling. When pressing **[-]** key, the display of sensors will be stopped.

4.7 Printing Data From Data Logger Card



While in STANDBY mode, if the **[Mean/Weight]** and **[alarm]** keys are pressed simultaneously, the data of the logger card can be transferred over the RS-232-port to a computer or an external printer. Only the data corresponding to the preset location-number will be sent.

4.8 Erasing Data From The Data Logger Card



Simultaneously pressing the **[Mean/Weight]** and **[Factor C]** will activate the erase data mode for the data storage card. After confirmation of the security fail-safe through pressing the **[+]** key, all data on the data storage card is erased.

This operation is only possible in the STANDBY mode.

4.9 Alarm and Error messages

When the alarm option is activated on the unit, any measurements exceeding the pre-selected threshold will be displayed on the LCD screen and be accompanied by an audible alarm. Other instrument errors or warnings such as low battery-capacity or high pump-motor current will also be displayed and will be accompanied by an audible alarm. These error messages will also be stored on the data storage card and sent over the RS-232-port to the PC.

4.10 Modem card use

Before start up the unit, latest while in STANDBY mode, enter the modem card and connect the phone line to the card.

Now start the GRIMM Software where you can dial up this phone line. The computer will establish the connection fully automatic (like an email). Than the instrument may be used as before .

	Do not use a combination card, such as a SEA net and analog modem.
	The modem card can call of from the basic 80 Kbytes data on request. This can be done daily 8or even on-line9 More details se SOFTWARE Manual !

4.11 Emails via the dust monitor

While in STANDBY mode, press the **[+]** key and the display will show the first line from the 1st message received. If you want to continue to read ,simply press the **[-]** key.

	Emails to the instrument can be send via GRIMM Software to the dust monitor., but not from the dust monitor to the computer

4.7 Short overview for all key functions

Certain key functions are just in the RUN-Mode or in the STANDBY-Mode activated. The [and] in the "key"-column means that two keys are simultaneously pressed.

key	Function-display	RUN-Mode	STANDBY
ON / OFF	Power up and power down the instrument with battery or power supply.		
Mean / Weight	The average collected dust mass in µg/m3 and the sample volume will be displayed.	X	
Mean / Weight [and] **+**	Average of all measurements after filter change and sample volume will be displayed.	X	
Mean / Weight [and] **-**	Backup the averages and sample volume.		X
Standby	Stop the active measurement.	X	
Standby	Start a new measurement.		X
Standby	By question „Filter Changed?": display the serial number and working hours.		
+ or **–** (just model 1.108 and 1.109 from Version 8.60)	Changing the storage interval: After put in the storage card the interval can be changed with [+] or [–] on the LCD.		X
Size Mode	Fix the current values on the LCD	X	
Size Mode [and] **+**	Select the measuring mode: change mass concentration [µg/m^3] in particle concentration [P/l] or reverse.		X
Battery / Location	Display the current battery capacity Display the location number (1...99)	X	X
Battery / Location [and] **+**	Display the current battery capacity Change the location number above	X	X
Alarm	Scan the adjusted alarm value	X	
Alarm [and] **+**	Increase alarm thresholds		X
Alarm [and] **-**	Decrease alarm thresholds		X
Date / Time	Display the date and time	X	
Date / Time	Short press: display Long press: adjust date/time		X
Factor C	Display the adjusted gravimetric factor	X	X
Factor C [and] **+**	Increase gravimetric factor		X
Factor C [and] **-**	Decrease gravimetric factor		X
+	Periodic display of the analog input values (sensors) will be activated	X	
-	Periodic display of the analog input values (sensors) will be deactivated	X	
Mean / Weight [and] **Factor C**	Old storage dates will be completely deleted. Storage card will be formatted.∗		X

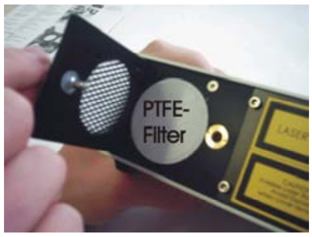
∗Every dust monitor accept just formatted storage cards. After buying a new card or change of a dust monitor to another the storage card have to be formatted.

5 Measuring operation

5.1 Start-up

NEVER START ANY UNIT WITHOUT **PTFE-FILTER**. FAILURE TO DO SO WILL CAUSE PERMANENT DAMAGE TO THE PUMP!

Verify that the filter is properly placed in the filter-holder. If a gravimetric analysis is to be performed on the filter, a “clean filter” weight must be determined prior to the start on any measurement. For more details see section 5.6.



Remove the black protection-cap over the air inlet before starting the instrument (see figure 1-**1**).

Do not attempt to remove the blue plastic air inlet connector.

The blue plastic air inlet connector is the attachment mechanism for the optional sampling heads.

Connect the power supply (see figure 1-**2**) or insert a charged battery into the battery slot. Press the black battery support button when inserting the battery into the compartment. The battery must be completely inserted into the compartment for a proper connection to be established. The user should hear an audible “click” as the connection is established. The battery is removed by pressing the battery support button. This will allow the battery to slide out of the battery compartment approx. 3-5 mm.

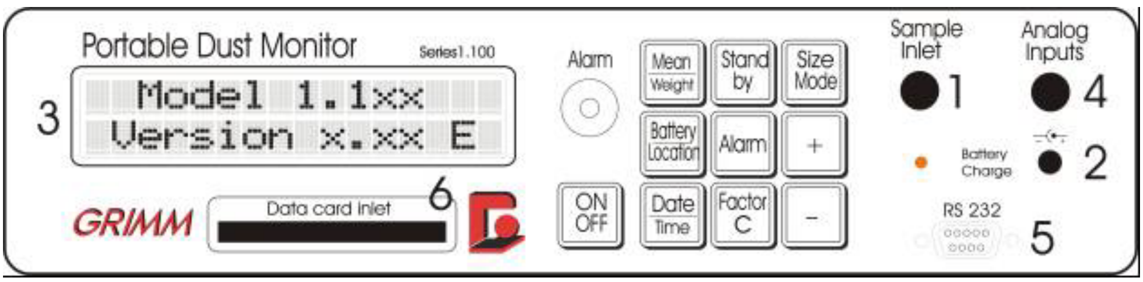


figure 1: front panel model 1.100

When you press the [ON-OFF] key once the unit is powered up, the LCD (figure 1-**3** previous page) will display the “model-number” and the “firmware version” installed on the unit.

„MODEL 1.1XX
VERSION 8.XX E“

Approximately 10 seconds later the unit will display the date and time.

Date	**28. 8. 5**
Time	**9:30:12**

In addition the units will display the data logger card size and the card version of the last card used,

Card	**512 kByte**
Version	**8.60**

the selected data storage interval (normally 1 min) and finally the available data storage capacity in day, hours and minutes in this mode.

Interval:	**1min**
free:	**5d18h40min**

This complete cycle takes about 20 seconds until he display shows the last question

filter changed ?
press +:yes -:no

By pressing now the [Mean weight] key will allow the user to view the **theoretical filter weight** and the sampled air.

Weight	**x,xx**	**µg**
Volume	**x,xxx**	**m³**

The unit will now prompt the user as to whether the gravimetric-filter should be exchanged. If you answer with yes (by pressing the [+]-key), the calculated filter weight as well as the affiliated sample air-volume will be reset to zero.

Weight	**0,00**	**µg**
Volume	**0,000**	**m³**

By pressing the [-]-key, by the question the instrument continues accumulating the mass and volume data. In order to prevent an inadvertent erasure of the stored values, the user must press and hold this key for at least 1 second. The operation will be confirmed by an audible “beep”. The instrument will then initiate a system self-test and ZERO calibration check.

Self Test

This process will last for approximately 30 seconds.

If the user wishes to modify any of the operation parameters during this time, the instrument must be placed in the STANDBY mode.

Standby Mode
Press 2nd Key

Returning the instrument to the measurement mode will automatically reactivate the system self-test and calibration check. The system self-test and calibration consist of a series of electrical diagnostics and measurements performed with internal “clean” sheath air. When the system self-test and calibration are complete, the display will read

Self Test OK

If an error should arise with the self test, then the message appears

New Self Test

If this news should appear more than once, then the massage appears

Fatal Error
Please Check

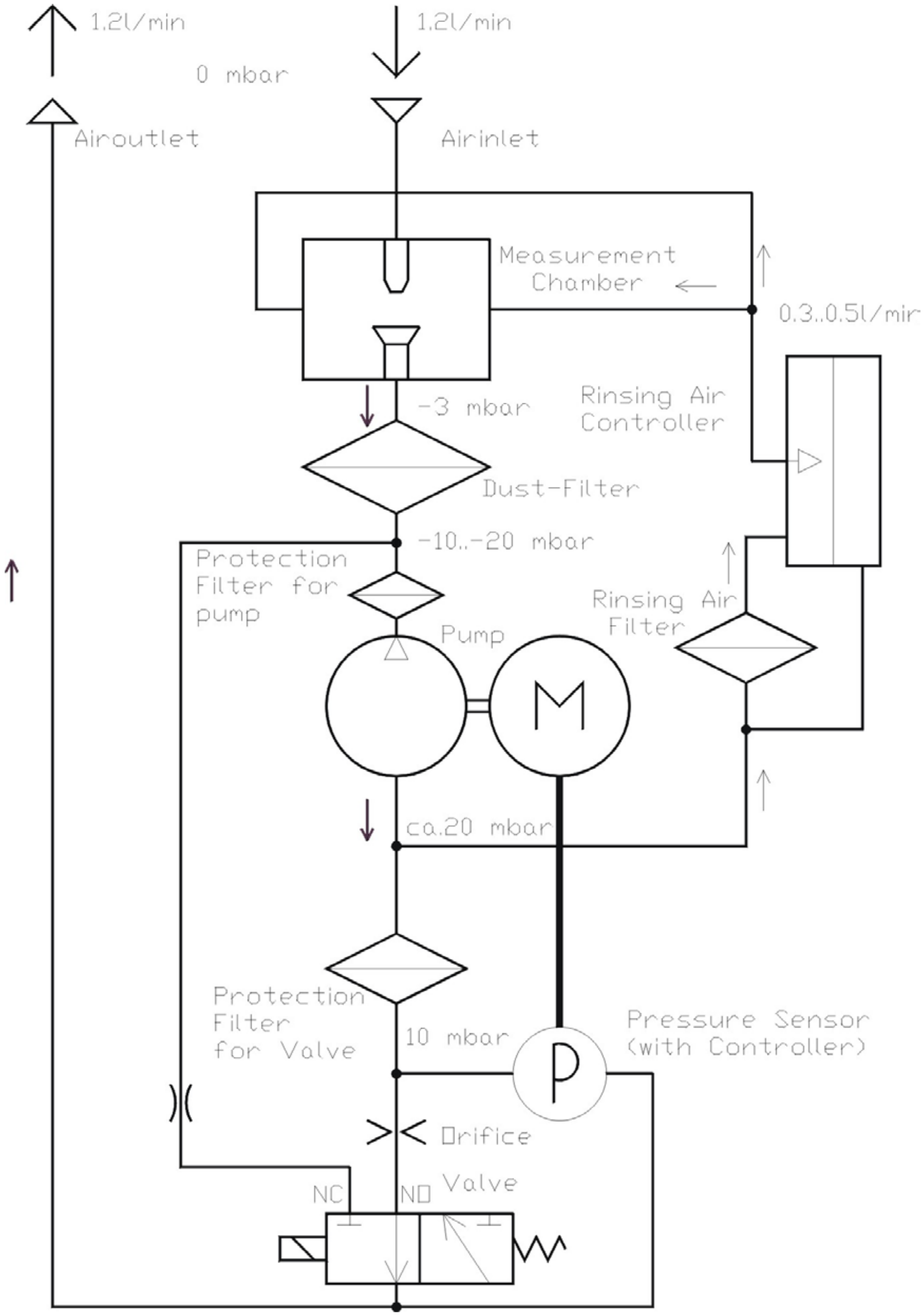
and an appliance-disturbance is existing so, with must be remedied. Possible causes for this are:

- a lint, that hangs at the air-nozzle and sticks out into the laser-ray,
- very unclean measurement cell
- an instrument-temperature over 50°C
- a laser or another hardware-defect.

Try first to remedy the disturbance through a thorough cleaning of the air inlet. (See also chapter 10.3 Cleaning of the measuring chamber

. Alternatively, the instrument must be send to your nearest representative for service.

Pneumatic of the Models 1.100



5.2 Measurement

Each measurement is started with a self-test, with which the measurement-cell is rinsed through pure-sheathed air. Several different internal measurements are enforced with it, from which conclusions of the status of the instrument are made. If all tests pass, the news appears on the display **“Calibration OK“** and measurements will be shown, such as:

>.30µ	**54587 /l**
>2.0µ	**290 /l**

After 6 seconds the collected results will appear on the LCD screen. Measured results are updated every 6 seconds. After the first minute, the collected values displayed will be reported on a 6-second sliding average.

5.3 Data Presentation on the LCD

In the standby-mode, the user may pre-select the desired measurement mode. The user may select the desired mode through pressing the **[Size mode]** and **[+]** or **[-]** key at the same time.

This process will activate the internal select measurement mode. In this mode the user will have the ability to view all of the associated measurement options. These options are advanced across the LCD screen via the **[+]** and **[-]** keys. Depending on the model of the dust monitor various measurement modes are available.

5.4 Factory calibration

Every unit is in our production with NIST-(National Institute of Standards and Technology) certificated, monodispers Latex on the respective size channels calibrated. Afterwards the unit is calibrated to a “mother unit”. The acceptable control limit is +/- 5%. The Calibration Certificate is in the transport case enclosed.

Once annually the calibration condition of the units must be checked. For this the unit must be sent to the manufacturer. There the candidate is checked with a “Mother unit”. You can get from the manufacturer if necessary a calibration tower and a “Mother unit” (reference unit). For operation of the calibration system a person trained by the manufacturer is to be used. The “Mother unit” must be sent annually to the manufacturer, there it will be checked with monodispers Latex aerosol and certified.

All are factory calibrated to ambient conditions and any correlation check is ONLY recommended when a different aerosol is used.

5.5 Gravimetric Factor

The particle size analyzer/dust monitor determines the dust-concentration (counts/liter) through the optical-light-scattering method directly; however, the mass concentration is determined by extrapolation. The calculated mass concentration may be corrected to the specific aerosol measured with the gravimetric- correction factor (C-Factor). Since the entire sample measured by the monitor is collected on a filter, this value can be quickly determinate.

The C-Factor is dependent on the

- density,
- refractive index and the
- form of the aerosol

collected and should be established with each different aerosol. This post correction is the most accurate manner to adjust the instrument to the specific aerosol measured.

- Each instrument is factory calibrated for size and mass as described in section 5.4 and leaves the factory with a C-Factor = 1.0.
- To avoid any data modifications, all data collected is stored on memo card and PC with the C-Factor = 1.0.

Determination of the gravimetric-factor (C-Factor)

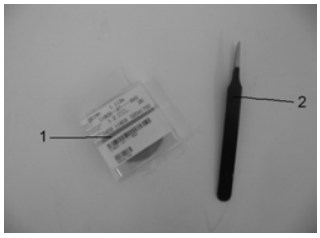
Hints:

- This procedure requires a four decimal place analytical balance.
- A minimum of 1mg of particulate must be collected on the filter to ensure optimum precision.
- A clean filter weight must be obtained prior to the initial aerosol sampling. This weight will be recorded and applied later during the final calculations.

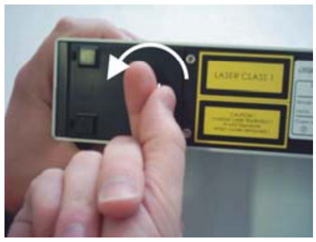
Procedure:

① Filter
Use the white filter, not the blue material in the filter box!

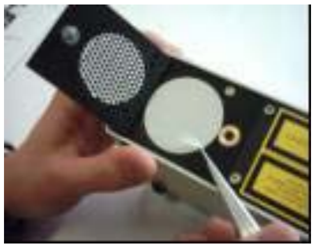
② Tweezers for filter (very fine surface) with this tweezers you can't damage the surface of the filter.



1 Open the filter chamber door



2 Remove the old filter with tweezers from the filter-holder assembly.



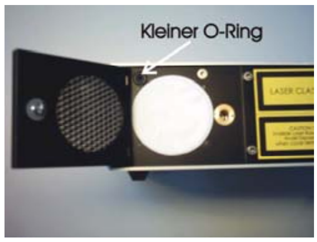
3 Using a cotton swab, thoroughly clean the inner walls of the filter housing.

4 Visually inspect this area to ensure that all particulate matter is removed.

5 Next, clean the sample inlet as outline in section 9.3

6 Weigh and record the weight of a new filter. For the highest level of confidence and accuracy, weigh the new filter at least three (3) times and record all weights. The average weight from a minimum of three (3) weights should be used in the final calculations.

7 Place the new, pre-weighed filter in the filter housing. Make sure the filter over laps the "O" ring completely. Is the filter in the right position you will see 4 points around the filter.
Use extreme caution when handling the filter and never touch it with your fingers!
Always use pre-cleaned tweezers. Failure to do so may add weight to the filter causing error in the calculated C-Factor.



8 Carefully close the filter-holder and tighten the screw firmly.

9 Power up the unit. When the instrument prompts the user with the question "Filter Changed?" reply yes by pressing the [+]- key. This process resets the calculated filter-weight to zero.

10 Make measurements of the aerosol for which the new C-Factor is being determined. While the user can interrupt the testing procedure as many times as they desire during the determination of a new C-Factor, the prompt "Filter Changed?" must always be answered with no (the [-] key). The instrument will continue to apply the total weight collected to the test series until these values are reset to zero.

11 The current calculated filter-weight may be viewed on the LCD screen by simultaneously pressing the [Mean/Weight] and [+]-key or on the PC via the RS-232-port.

12 After collecting a sufficient amount of particulate matter on the filter, the test series may be stopped. Place the instrument in the STANDBY mode and turn the instrument off.

13 Carefully remove the filter, taking care that none of the particulate on the filter is disturbed. Place the filter in a desiccate chamber for twenty-four (24) hours. Take the filter out and weigh it at least three (3) times. The difference between the clean filter weight and this measurement is the collected dust mass.

14 The gravimetric-factor may be calculated using this equation

$$C-Factor = \frac{Filter\ \ weight\ \ (loaded\ \ filter - clean\ \ filter)}{Theoretica\ l\ weight\ \ shown\ \ by\ the\ instrument}$$

or software provided with the unit. To improve the precision and accuracy of the unit, regular verification of the C-Factor is recommended.

Under normal conditions this value should not vary in ambient air by more than +/- 30% of the factory set-value. However, widely varying values can be exhibited with heavier aerosols such as metal powders. The calculated C-Factor can now be entered into the instrument. All dust mass concentration values as well as their mean values are multiplied with this factor and displayed on the LCD. However, the collected data on the data storage-card and data transferred via the RS-232 interface are not corrected using the C-Factor. The established C-Factor for the subsequent measurements is transferred in the data file and can be applied to the data downloaded to the PC. The C-Factor can then be applied to the collected data at a later time.

Use of the Cyclone Pre-separator

During the measurement process, particles categorized as being larger than the largest size channel can only be categorized as such (i.e. particles classified with diameter > 20 um will be reported as such regardless of how much larger actually are). Therefore, the potential exists to under estimate the mass of larger particles.

As a rule, in ambient air large particles are normally rare and errors associated with the mass analysis can be considered as negligible. However, with the measurement of very coarse aerosols, this error increases considerably and will affect the final total dust concentration and the calculated filter weight. It is recommended that when analyzing very coarse aerosols, a cyclone adapter be attached to the sampling path. The use of the cyclone will eliminate particles above the size of specified. The cyclone employs a pre-collection bag so that the coarse particles can be weighed and evaluated at a later time.

6 RS-232-Port

6.1 Assembly

The serial interface consists of the 9-pin connection on the front panel of the instrument and is marked "RS-232 ". Using this interface, the instrument can be operated and the collected data transmitted to an external printer or a PC.

The RS-232 port is used to establish the parameters for the internal alarm system. Due to the unique configuration requirements for this connection, the cable employed for this operation must be GRIMM part number 1.143 E serial connection cable.

<u>The pin-configuration is as follows:</u>

Front panel

Pin #2: RxD	(Data reception)
Pin #3: TxD	(Data transmission)
Pin #5: GND	(Signal-ground)
Pin #7: Alarm 12V/0.5A	(signal current for optional alarm-unit)
Pin #8: CTR	(Start up contact)

6.2 Transfer-protocol

The baud rate for normal data transfer is **9.6 kbds**. **8 data-bits without parity** and **a stop** bit are used. The software protocol is supported as **XON/XOFF**. Real time data transfer is via the **CTS cable**. Maximum baud rate can equal **57.6 kps**. Data-transfer to a printer in an ASCII format is possible through simultaneous pressing of the [Mean/Weight] and [alarm] – key.

6.3 RS-232 Commands

Some commands may be sent to the instrument via the RS-232-port. In turn, the instrument will confirm the receipt of these commands through an echo. Capital or lower case letters are used to initiate the commands. Numerical values can only be altered in the standby-mode. Commands must be confirmed with a CR (Carriage Return) or through pressing the <Enter>. For example, to enter the program to alter values for the alarm settings, the user would type the command then either use the <Enter> or “CR” key.
This process is accessible using any terminal program that supports the **XON/XOFF** protocol such as the the TERMINAL program in Microsoft Windows^{TM} .
The following keys correspond to specific settings for access using the MS Terminal program. The function-keys can be programmed for specific commands to the instrument.

A Current alarm-value. Altered in the standby-mode.

A
Alarm : 0 /l :

B Current battery-capacity

B
Battery Power : 11 %

C Count mode switched on / mass-concentration turned off (only available in software Version 5.X and 6.X)

D ASCII-data-transfer of the data-storage-card (only accessible in the standby)

D
Memocard : 12.30 from: 9G040001
Location : 2 :
P: 5 2 23 14 28 2 0 0 44 13 65 185 0 0 0 1

K: 427 800 1611 0 0 35 91 0 0 0 0
P: 5 2 23 14 33 2 20 0 42 30 64 185 0 0 0 1

C_: 420648 312631 253253 138895 73489 33731 13396 6428
C_; 3061 2279 1395 647 493 365 274 197
c_: 197 146 110 77 36 18 10 7
c_; 4 2 0 0 0 0 0 0

P: 5 2 23 14 56 2 0 0 41 20 64 185 0 0 0 1
K: 546 825 1631 0 0 35 91 0 0 0 0

E Sends error code

Error Code:	**LCD**	**Explanation**
"128"	NEW SELFTEST	fault self test
"64"	NO MEMOCARD	Also with U_{mc} < 2.8 VX or wrong version or card with dates from another unit
"32"	CHECK NOZZLE	Whirls >5%
"16"	NO OPERATION	Battery empty = 0%
"8"	PLEASE RECHARGE	Akkukapazität < 10%
"4"	PUMP CURRENT TOO HIGH	I_{mot} > 100%
"3"	FLOW-ERROR	Pump-regulation outside of range
"2"	CHECK FILTER	I_{mot} < 20%
"1"	CHECK FILTER	I_{mot} > 60%

F Measurement mode switches on. Results will be received every 6 seconds

G Select gravimetric factor. This value can only be altered in the STANDBY mode. The selection range is between 0.1 to 9.9 in incremental of 0.05.

H Total operating hours

I Interval for normal output and the storage card data output. In Standby-Mode it can be changed.

Interval adjustment of the storage:

0	=	*1*	*minute*
1	=	*5*	*minutes*
2	=	*10*	*minutes*
3	=	*15*	*minutes*
4	=	*30*	*minutes*
5	=	*60*	*minutes*

J Output of the channel thresholds in µm (only in the count mode)

J *channel thresholds*

Jc:	*0.25*	*0.28*	*0.30*	*0.35*	*0.40*	*0.45*	*0.50*	*0.58*
Jc;	*0.65*	*0.70*	*0.80*	*1.00*	*1.30*	*1.60*	*2.00*	*2.50*
jc:	*2.5*	*3.0*	*3.5*	*4.0*	*5.0*	*6.5*	*7.5*	*8.5*
jc;	*10.0*	*12.5*	*15.0*	*17.5*	*20.0*	*25.0*	*30.0*	*32.0*

number of particle

C8:	*43167*	*31404*	*25011*	*13215*	*6935*	*3095*	*1290*	*665*
C8;	*370*	*300*	*180*	*55*	*45*	*40*	*35*	*27*
c8:	*27*	*11*	*8*	*3*	*2*	*2*	*0*	*0*
c8;	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*

L Location number. Can only be altered in the standby-mode.

M Median values and total volume of air sampled

M

Mc:	*447430*	*324824*	*258999*	*136537*	*70110*	*31757*	*12825*	*6485*
Mc;	*3357*	*2574*	*1640*	*847*	*648*	*493*	*366*	*269*
mc:	*269*	*201*	*154*	*114*	*58*	*32*	*22*	*16*
mc;	*8*	*3*	*1*	*0*	*0*	*0*	*0*	*0*

V: 0.0288 m3

O View capacity of data storage card (only accessible in the standby-mode). In this mode the user can erase all data on the data storage card by selecting the lower case “a” followed by the “ + “ key and then the “CR” or “enter” Key.

Q Quick-data-transfer of the data on the data storage card (only in standby-mode). If the data-transfer takes place via binary-representation the transfer rate can be as quick as 57.6K baud.

R Run. Start of the measurements from the STANDBY mode

S Switches unit in STANDBY mode
T Edit Time. Clock can be reset in STANDBY mode.
U Keyboard access
- **U=0** Unlocks the keyboard
- **U=1** Disables STANDBY mode.
- **U=2** Keyboard-access not possible

V Displays the version number of the monitor's hardware
V
Version : 12.30 E

W Shows software's calculated filter weight and associated sampled volume of air.
W
Weight : 2.4 ug Volume : 0.038 m3

Z Re-Zero mode. Resets air sample's mean values and total volume to zero.

! Shows model and version number of monitor
!
Model 1.109 Version 12.30 E

@ View the monitor's serial number
@
Ser.No. 9G040001

^A Transmits calibration factor of the analog-sensors

^B Set of baud rate for quick-data transfer
0 = 9.600 Baud
1 = 19.200 Baud
2 = 57.600 Baud

^D Interrupts the measurement data transmission

^E Starts up the measurement transfer / turns off of the fast-mode

^L Area/Country-date adjustment (E or U) (only in the standby-mode)

^T Timer operation for automatic on- and off form the dust monitor (from unit version X.40)

^Y Switch off the unit

Long Break Switch on the unit (when it´s with command ^Y switched off)

_ (Underline) output of „User"- text and analog input factors (only in the standby-mode)

Example:

_
$(1..4):Input 1: . V |Input 2: . V |Input 3: . V |Pressure hPa|
**(1..4): 1 | 1 | 1 | 179.3722 |` 0.000 V | 0.000 V | 0.000 V | 1.880 V |*

In the first line, the three "User"-selectable phrases are presented. In the second line, the three multiplication-factors for the analogous-signals and the offset values are shown.

$ Selection or change of the "User"-selectable phrases as well as factors for the analogue signal to the LCD-display. (Only in the standby-mode). Symbols (over ASCII 127) cannot be inputted. With the edition, the sign becomes ' °'(ASCII 248) as ' _' spent. Pre-determined texts can be selected with the TAB that through the enter key is confirmed. They can be headed also with any texts. The number-edition is always in five digits and begins from the 9. Position. If signs are entered, besides a decimal-point, an acoustic warning in generated. The multiplication-factor, that is referential on 1V, can then after the text be inputted. He can be changed through the command ` * ` alters.

Example:

Temperature-sensor	*0°C:=:3.0V and 50°C:=:8.0V*
User-Text:	*Temp.: . °C*
User-Factor:	*10.0 [°C/V]*
Offset:	*- 3.00V*

***** User-fractions alter (only in the Standby-mode)

? Hilfe für Befehle

```
################### Help for Dust Monitor ###########################
|  A' Alarm                           | ^L' Land (for Date)    [Standby]|
| ^B Baudrate (Memocard)    [Standby]|  L' Location Code               |
|  B' Battery                         |  M' Mean Value                  |
|  D' Data Memocard        [Standby] |  O' Clear Memocard     [Standby]|
| ^D' Disable Output                  |  P' Preferences Modem  [Standby]|
|  E' Error                           |  R' Run Measurement             |
| ^E' Enable Output                   |  S' Standby Modus               |
|  F' fast                            | ^T Timer Set           [Standby]|
|  G' Gravimetry C-Factor             |  T' Time Set           [Standby]|
| ^G' Byte / Interval                 |  %' Memo free          [Standby]|
|  H' Runtime hours                   |  U' Unlock Keys        [Standby]|
|  I' Interval                        |  V' Version                     |
|  J' Output Channels                 |  W' Weight                      |
|  @' Serial-No.                      | ^Y' Power OFF                   |
|  $' User Strings (Analog Inputs)    |  Long Break: Power ON           |
|  *' User Factors (Analog Inputs)    |  Z' Zero Clear Mean             |
|  _' Output User Strings + Factors   |  !' Output Model + Version      |
#######################################################################
```

6.4 RS-232 data transmission

All data given are only in the „Count“ mode and only available in „Fast-mode“ (6 sec.) The data in the ‘P‘-Line are as following:

P-Line:

Year Mon Day Hr Min Loc GF Err Qbatt Im UeL Ue4 Ue3 Ue2 Ue1 I

P 0 8 17 10 17 8 0 0 130 6 0 0 0 0 0 0

The data in the “K” String are only needed for the service:

K-Line:

K_ 1794 4963 5422 0 0 0 0 0 0 0 0 0

The main line are the data obtained in the C = “ Count”-mode. (The example shown below was with a leaking filter obtained , this is not normal, but the values can be seen)

The channel size in µm can be seen with the command ‘J‘ and are :

J: 0.30 0.40 0.50 0.65 0.80 1.00 1.60 2.00
C0 20 5 5 0 0 0 0 0

j: 2.0 3.0 4.0 5.0 7.5 10.0 15.0 20.0
c0 0 0 0 0 0 0 0 0

C1 0 0 0 0 0 0 0 0
c1 0 0 0 0 0 0 0 0
.
.
c8 0 0 0 0 0 0 0 0
C9 10 0 0 0 0 0 0 0
c9 0 0 0 0 0 0 0 0
P 0 8 17 10 18 8 20 0 130 39 0 0 0 0

The Software variants 5.x up to version 8.x differentiate only in the amount of size channels. The Version 1 and 4 however has no count channel mode.

With the software version 6.x and 7.x are only the mass concentration values obtained, but in a resolution of 0.1 µg/m3, and therefore with a multiplication of 10! Starting from the version x.4.0 is the comma included Results are transmitted each minute with the pre-data as mentioned above.

The measurement data are indicated with "N_" for mass or "C_" of Counts. At the mass mode are the data in the pre-selected time interval transmitted. The data consequence is hereby as following: Inhalable, Thoracic, Alveoli and than PM-10. (Version 5.X)

With the Version 4.2 upwards is in addition the concentration value of PM-2,5 given

With the Version 5.X upwards is in the cumulative mode the size channels 1 to 8.

With the Version 6.1 is the sequence as following:

Total, Alveoli, PM-2,5, PM-1,0 and than the accumulative size channels 1 to 8. In the Count-mode are the accumulative channel sum from 1 to 8 in each minute given. Because of the fixed sample flow of 1.2 Litre per minute are this values corrected by 1/1,2 (= 0,83333..) and the counts/litre and given in full numbers.

Version : 5.0

P 5 2 7 13 47 4 0 0 130 17 0 1 0 0 0
K 53 1154 2833 224 1 0 20 51 4 0 0 0
P 5 2 7 13 48 4 20 0 130 42 64 1 0 0 0
N_ 13 12 10 10 9 7 4 2 1 1 1 0
P 5 2 7 13 49 4 20 0 130 41 64 1 0 0 0
N_ 14 13 10 10 11 8 6 4 3 2 1 0

P 5 2 7 13 55 4 20 0 130 40 64 1 0 0 0
C_ 9690 1713 222 44 9 4 4 1
P 5 2 7 13 56 4 20 0 130 40 0 1 0 0 0
C_ 9732 1783 233 49 13 10 4 0

In the Fast-mode are the results all 6 sec. transmitted as following:

P 5 2 7 13 57 4 20 0 130 40 0 1 0 0 0
C 1 269 83 24 9 2 1 1 1
C 2 231 81 17 5 0 0 0 0
C 3 239 71 21 8 4 4 0 0

P 5 2 7 13 58 4 20 0 130 40 0 1 0 0 0
N 1 38 14 9 9 35 32 29 28 28 28 28 28
N 2 9 9 9 9 6 4 2 0 0 0 0 0
N 3 11 11 9 9 8 6 4 2 1 0 0 0
N 4 13 13 10 9 10 7 5 4 3 0 0 0
N 5 23 17 11 11 19 16 14 12 11 10 10 0
N 6 49 19 10 10 45 43 41 39 38 38 38 28

Version : 4.0

P 5 2 7 15 8 4 0 0 130 15 64 1 0 0 0
K 53 1153 2836 225 1
P 5 2 7 15 9 4 20 0 130 40 64 1 0 0 0
N_ 34 20 12 12 0

The pre set of data indicated with a "P" correspond with the Bytes, stored on the data Memo card. The signal values (of the external sensors) and the Cal-Values at in 10 Bit, so that the Bytes, indicated with Ue1…Ue3 represent only the higher values. The low value Bits are combined in the UeL. The Bits 7 and 6 represent the Cal, the Bits 1 and 0 the Ue1. The max. value is 10 V, so the 10 Bit-Value needs to be multiplied with the factor 9.776E-3, if the signal is needed in Volt. The gravimetric factor is with the Version 7.x with the factor 100 and with all the other Versions with the factor 20 multiplied, this gives a resolution of 1% rep…5%. The battery capacity and the motor current are given n in percent. The data are as following:

Year Mon Day Hr Min Loc GF Err Qbatt Im UeL Kal Ue3 Ue2 Ue1 I

Explanation of the data in full mentioned above:

Year - Month - Day - Hour - Minute – Measurement Location (1..99) – gravimetric factor – Error code Battery capacity – Motor current – ZERO-calibration - Sensor 1 to 3 – Interval

After the Pre-data set (P) are the measurement data (N) for Mass or (C) for Counts given. With the mass values are also the mean values of the data transmission time displayed.

Conclusion:

The versions 6.x and 7.x are given as mass concentration values with a resolution of 0,1µm/m3 and the multiplication of 10! This is shown as comma and displayed as 3rd symbol the results.

1.104: Inhalable, Thoracic, Alveolar, PM-10, PM-2.5
1.105: Inhalable, Thoracic, Alveolar, PM-10, accumulative size channels 1..8
1.106: Total, Alveolar, PM-2.5, PM-1.0, accumulative size channels 1..8
1.107: PM-10, PM-2.5, PM-1.0
1.108 accumulative size channels 1..15 (only in the Count mode)

Specially with the model 1.108 are the results in counts given. Hereby means the "C" the counts in the submicron range and the "c" the counts above 2 µm for the larger particle (Multiplex-use).

Version : 7.40

P ' 5 2 16 9 30 2 100 192 130 39 128 1 0 0 0 0N_, 936 397 132

Version : 5.40

P 5 3 4 16 58 1 0 64 130 9 64 21 0 0 0 0K_ 1151 3165 4317 0 0 0 0 0 0 0 0 0
P 5 3 4 16 59 1 20 64 130 56 64 21 0 0 0 0
N_ 144 80 37 67 143 140 132 115 97 91 85 50

P 5 2 7 13 55 4 20 0 130 40 64 1 0 0 0 0
C_ 9690 1713 222 44 9 4 4 1
P 5 2 7 13 56 4 20 0 130 40 0 1 0 0 0 0
C_ 9732 1783 233 49 13 10 4 0

In the Fast-mode are the results all 6 sec. transmitted as following:

P 5 1 28 8 45 1 20 64 130 67 64 20 0 0 0 0
C0 88785 40485 6605 203 2 0 0 0 0
C1 95920 44045 7138 147 4 0 0 0 0
C2 103755 47615 7679 183 3 0 0 0 0

P 5 3 4 17 1 1 20 64 130 56 64 21 0 0 0 0
N0 56 56 38 51 55 52 44 24 11 0 0 0
N1 117 63 37 50 115 112 105 90 69 63 63 63
N2 74 61 33 56 73 70 63 47 34 23 23 0

Version : 4.40

P 5 3 4 17 2 1 20 64 130 9 64 21 0 0 0 0
N0 83 68 36 63 19
N1 120 99 44 91 21
N2 71 66 36 61 19

Version : 8.40

P 5 2 2 14 11 1 20 64 130 35 0 0 0 0 0 0
C0 20280 9400 5980 3600 2365 1290 735 375
c0 386 87 18 4 2 1 0 0
C1 19795 9405 6035 3500 2325 1355 785 485
c1 430 115 19 2 1 1 1 1

7 GESYTEC Instruction Section

(This is special German data transmission protocol mode)

7.1 Change from Standard-mode to GESYTEC-mode

The change from the **Standard** to the **GESYTEC-mode** is only possible via the keyboard of the dust monitor. Simply switch on the instrument by pressing ON/OFF and „+"- key at the same time, and keep both key pressed unit on the LCD-Display the model and firmware version is shown, after that the instrument will directly change over to the GESYTEC-mode automatically. This is shown on the display with an extra „S" at the end.

To return to the Standard-mode start the instrument by pressing the „ON/OFF" and the „-"- key at the same time. Now the instrument return to the pre-selected mode, in Europe as „E" or in America as„US" version.

7.2 Settings in the STANDARD mode

The only option is an additional message in the RS-232-string „:" for the Gesytec- pre-setting. During the Standby mode is an extra signal as „:" (ASCII-Code 58) transmitted, with will open the sub menu as following:

:

--Settings for the Gesytec--

: Instrument identification 123

Serial -number : 987

Baud rate RS232: 9600

[K] [S] [B] [Esc] [ENTER] ?_

After transmitting the „K" is it possible to change the Instrument identification . Possible are the ranges from 0 to 255. The Gesytec- Serial number can after transmitting the „S" be set from 0 to 999. After sending the signal „B" will be the following helping text be shown. Now a number between 0 and 4 will be expected:

Baud input : 0=1200 : 1=2400 : 2=4800 : 3=9600 : 4=19200

Baud rate RS232: 9600 ?_

By pressing the Escape key it is possible to exit from this submenu, without changing the settings By pressing the [ENTER] (ASCII-Code 13) are the change stored.

7.3 GESYTEC- Operation

Use directly from the instrument keys

The instrument can be operated independently from the pre-selected mode with two exemptions:

1. The Gesytec- Instrument recognition and the selected Baud rate can be seen by pressing the „Size-Mode“- key in the standby mode.
2. The question if the filter needs to be exchanged remains only for ca. 3 Sec. On the LCD display, after it changes automatically in the Standby mode.

Operating the unit via an external PC

This is the normal use of the protocol, since the instrument is in an environmental network installed and transmits the results when asked only (GESYTEC-Protocol).

All Standard-mode operations can NOT be used when in this special mode

The signal come as „ST“- Telegram. The following „ST“- Telegram s are realised:

„**R**“ = (Run)
„**S**“ = (Standby) → Stop of measurement
„**N**“ = (Null) (acts like in Run)

The „ST“- Telegram can be selected as ASCII- or Hex- instructor and are in the following ASCII-Format:

<u>ASCII-Format:</u>

Field-Nr.	Byte-Nr.	Data format	Field description
1	1	<STX>	Start of Text
2	2-3	ST	Telegram code
3	4-6	nnn	Instrument code
4	7-8	#A	Empty space key, (mostly a letter)
5	9	<ETX>	End of text
6	10	<BCC1>	Block sign lower nibble
7	11	<BCC2>	Block sign upper nibble

<u>Hex-Format</u>:

Field-Nr.	Byte-Nr.	Data format	Field description
1	1	<STX>	Start of text
2	2-3	ST	Telegram code
3	4-6	nnn	Instrument code
4	7-8	hh	upper nibble, lower nibble
5	9	<ETX>	End of text
6	10	<BCC1>	Block sign of the upper nibble
7	11	<BCC2>	Block sign of the lower nibble

The data obtained, with is only generated when the request „DA“ was entered, is identified as „MD“-Telegram, in the following sequence and Format:

Field-Nr.	Byte-Nr.	Data format	Field description
1	1	<STX>	Start of Text
2	2-3	MD	Telegram code
3	4-6	03#	Amount of measuring places
4	7-10	nnn#	Instrument identification
5	11-19	±nnnn±ee#	Value of PM10
6	20-22	hh#	Operational status
7	23-25	hh#	Error status
8	26-29	nnn#	Serial number
9	30-36	hhhhhh#	Temperature in 1/10°C (F= Minus)
10	37-40	nnn#	Instrument identification + 1
11	41-49	±nnnn±ee#	Value of PM2.5
12	50-52	hh#	Operational status
13	53-55	hh#	Error message
14	56-59	nnn#	Serial number
15	60-66	hhhhhh#	relative Humidity in 1/10%
16	67-70	nnn#	Instrument identification + 2
17	71-79	±nnnn±ee#	Value of PM1
18	80-82	hh#	Operational status
19	83-85	hh#	Erros status
20	86-89	nnn#	Serial number
21	90-96	hhhhhh#	Motor current in %
22	97	<ETX>	End of Text
23	98	<BCC1>	Block symbol lower nibble
24	99	<BCC2>	Block symbol upper nibble

The „DA“- Telegram must contain the following format :

Field-Nr.	Byte-Nr.	Data format	Field description
1	1	<STX>	Start of text
2	2-3	DA	Telegram code
3	4	<ETX>	End of text
4	5	<BCC1>	Block sign of the upper nibble
5	6	<BCC2>	Block sign of the lower nibble

The sign „#“ symbols the space key, „nnn“ the numbers, „ee“ s the exponent and „hh“ is the hexadecimal presentation of one Bytes.

The block diagram is composed, by adding bytes over all transmitted symbols (incl. STX and EXT) and the Exclusiv-Or-Sum (staring from Zero). The result is shown as two nibble hexadecimal signal and transmitted.

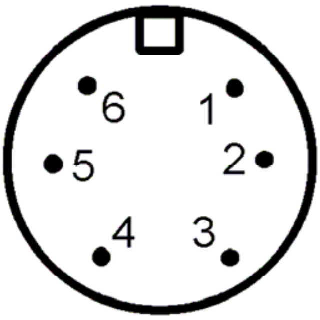
The error status is presented as Error-Code from the dust monitor. The Bit 1 in the operational-status- is set as Byte, as long as the dust monitor is in the Standby-mode. The Bit 2 is send during the self test phase only.

8 Sensors

8.1 Analog input

The instrument is equipped with an analog connection. The connection is a 6-pin socket capable of receiving signals from three individual inputs. The socket is located on the right upper corner of the front panel. The input signal to be used this connection is between 0 and 10 volts for each signal. The connection-socket can also supply the external power for 10 volts up to 40 MA.

Sensors that are most often used are temperature, air velocity, barometric pressure, humidity, CO2, etc. The measured analog values are stored on the data storage card and can be shown on the LCD display. The resolution amounts to 10 bits (approximately 10mV). The text and factors for the LCD display are adjustable.



①	Input 1
②	Input 2
③	Input 3
④	GND (mass)
⑤	+10V/40mA
⑥	(Free)

Front view: analog socket pin assignment

8.2 Barometer (intern)

In the newer units is a barometric sensor integrated. His values are stored and issued as Ue4. On the LCD the value is shown as **„pressure“.** For display the sensor measurements on the LCD see chapter 4.6.8 [+] and [-] Keys

8.3 Basic sensors

The Sensors for ambient temperature und humidity are standard installations. They are necessary for the proper operation. For display the sensor measurements on the LCD see chapter 4.6.8 [+] and [-] Keys

8.4 Weatherstation

By means of the optional "*one wire*" connector is it possible to connect additional climatic sensors for

① wind direction

② wind velocity and

③ rain

Moreover there is the possibility to integrate a PAH- Sensor (Modell 130). Additionally this sensors need more space on the storage card and decrease thus the storage time. To connect additionally sensors the unit must be switched off. After switching on the firmware version will automatically change. Connecting the rain sensor, for example, the version 7.80 will be switched in version 7.81. After connection the data storage must be erased. If operating without additionally sensors so the storage space will be for it reserved.

The additionally sensor can be reinstalled when the unit is

- switched on without additional sensors and afterwards delete the data storage.
- afterwards switch the unit off and on and check on the LCD the shown firmware version.

9 Data Storage Cards

9.1 Storage card capacity

The standard data storage card has 512 Kbytes capacity and can hold measurements for up to four days of continuous measurements. Data cards with larger storage capacity are also available. The maximum capacity is presently 6 MB. The data are stored in the procedure "first-in, first-out" (ring-buffers). This process ensures that when a storage card becomes full, the oldest data sets are overwritten first.

The data storage card must be installed before start up a measurement, otherwise data losses are impend.

The data storage card holder is the small black slit located on the front panel of the monitor.

To use a data storage card, insert the card with the imprinted arrow facing upwards.

The user should hear an audible "click" when the card is completely inserted and the card will protrude about 1 cm from the front of the monitor. All measured data will be automatically written to the data storage card.

A storage card will only be accepted of the unit when it is erased or in the same unit used and version and the number of additional sensor are not changed.

Before you start to work with a new card you need to format this card. See chapter 4.8 Erasing Data From The Data Logger Card.

The procedure is as following:

When pressing [Mean/Weight]- and the [Factor C]-key simultaneously the message

Clear Memorycard
Are you sure ? +

will show up. If you are sure you have to erase, press the [+]-key.

If no memo card is in the instrument the same procedure will clear the 80 kBytes internal memory.

When a data storage card is not used, the following message will appear on the LCD:

No Memorycard !

When starting up a measurement, a warning-tone, as a “beep”, will sound if the data storage card is not in place.

When a data storage card was previously used in another unit which was not the same model or operated in a wrong version, the following massage will appear on the LCD:

WRONG VERSION !
Clear Memorycard

You need to format this data storage card. See chapter 4.8 Erasing Data From The Data Logger Card.

9.2 Data preservation / backup battery

With each new memory card a new buffer battery is supplied separately in order to assure long battery life. First insert the battery into the memo card (a small black port on the lower side) before using the memo card.

The data storage card is equipped with it’s own backup battery. The lifetime of this battery is approximately over one year. When data storage card is not in use the battery should be removed from the data storage card. When the storage card is in the unit the power supply goes over the unit. Please read the stored data out before changing the backup battery, otherwise data losses are impend.

CAUTION:

If the data storage card being used was previously used in another unit which was not the same model, the card will be automatically re-formatted and all of the previously collected data erased.

The data storage card is only to be inserted or removed in the STANDBY mode. Failure to do so may cause all previously collected data to be erased.

If there is no memo card in the instrument, the data will be stored in the internal 80 Kbytes memory. This represents about 24 hours.

9.3 Storage Time

There are memory cards in different sizes available, ranging from 1 Mbyte to 6 Mbyte. Depending of the size of the memo card and the configured storage time interval, the maximum storage time will be as indicated in the spread sheet.

Table 1: *storage times of the data storage card for version 4.x0* (Y=year; d=day; h=hour)

INTERVAL	1min	5min	10min	15min	30min	60min
INTERN (80 Kbyte)	2d 4h	10d 22h	21d 21h	32d 19h	65d 15h	131d 6h
256 Kbyte	7d 0h	35d 0h	70d 0h	105d 0h	210d 0h	1Y 55d
512 Kbyte	14d 0h	70d 0h	140d 0h	210d 0h	1Y 55d	2Y 110d
1 Mbyte	28d 0h	140d 0h	280d 0h	1Y 55d	2Y 110d	4Y 220d
6 Mbyte	168d 0h	2Y 110d	4Y 220d	6Y 330d	13Y 295d	27Y 225d

Table 2: *storage times of the data storage card for version 5.x0*

INTERVAL	1min	5min	10min	15min	30min	60min
INTERN (80 Kbyte)	1d 11h	7d 7h	14d 14h	21d 21h	43d 18h	87d 12h
256 Kbyte	4d 16h	23d 8h	46d 16h	70d 0h	140d 0h	280d 0h
512 Kbyte	9d 8h	46d 16h	93d 8h	140d 0h	280d 0h	1Y 195d
1 Mbyte	18d 16h	93d 8h	186d 16h	280d 0h	1Y 195d	3Y 25d
6 Mbyte	112d 0h	1Y 195d	3Y 25d	4Y 220d	9Y 75d	18Y 150d

Table 3: *storage times of the data storage cards for version 8.x0*

INTERVAL	1min	5min	10min	15min	30min	60min	6sec	3sec	2sec	1sec
INTERN (80 Kbyte)	21h 4min	4d 12h	9d 0h	13d 13h	27d 2h	54d 4h	2h 10min	1h 5min	0h 43min	0h 21min
256 Kbyte	2d 21h	14d 10h	28d 21h	43d 8h	86d 16h	173d 8h	6h 56min	3h 28min	2h 18min	1h 9min
512 Kbyte	5d 18h	28d 21h	57d 18h	86d 16h	173d 8h	346d 16h	13h 52min	6h 56min	4h 37min	2h 18min
1 Mbyte	11d 13h	57d 18h	115d 13h	173d 8h	346d 16h	1Y 328d	1d 3h	13h 52min	9h 14min	4h 37min
6 Mbyte	69d 8h	346d 16h	1Y 328d	2Y 310d	5Y 255d	11Y 145d	6d 22h	3d 11h	2d 7h	1d 3h

Table 4: *storage times of the data storage card for version 9.x0*

INTERVAL	1min	5min	10min	15min	30min	60min	6sec	3sec	2sec	1sec
INTERN	1d 2h	5d 13h	11d 3h	16d 17h	33d 10h	66d 21h	2h 40min	1h 20min	0h 53min	0h 26min
256 Kbyte	3d 13h	17d 20h	35d 16h	53d 12h	107d 0h	214d 0h	8h 33min	4h 16min	2h 51min	1h 25min
512 Kbyte	7d 3h	35d 16h	71d 8h	107d 0h	214d 0h	1Y 63d	17h 7min	8h 33min	5h 42min	2h 51min
1 Mbyte	14d 6h	71d 8h	142d 16h	214d 0h	1Y 63d	2Y 126d	1d 10h	17h 7min	11h 24min	5h 42min
6 Mbyte	85d 14h	1Y 63d	2Y 126d	3Y 189d	7Y 13d	14Y 26d	8d 13h	4d 6h	2d 20h	1d 10h

Some important hints:

- When there are additional sensors among the sensors 1…4 attached to the dust monitor, the indicate storage time will decrease accordingly.
- If you are using a longer time interval, you have to wait till the time interval is finished, before stopping the measurement other wise you will loose data (if you had set the interval to 60 min and you stop the measurement short before the end of the interval, you will loose 59 min of measured data).
- When the time indicated in the spread sheet is exceeded, the oldest data on the memory card will be overwritten.

9.4 Write Protection

The memory card has a build in write protection, to prevent a undesired overwriting of measured data. The lever for the write protection is on the side of the memory card. If you want to store new data, you must remove the write protection by sliding. If you want to secure stored data you activate the write protection.

9.5 Modem use

Since version F of the 1.108 (from 2000) is it possible to use the new type of PCMCIA cards with larger storage capacity and the ability to slid instead of the data logger card the MODEM CARC into the dust monitor. This analog cards work with most telephone system in the world, but no warranty for proper function can be given by the manufacturer.

<u>The procedure is simple:</u>

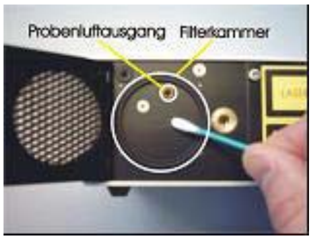
- Enter the modem card into the data card inlet
- Connect the telephone line to the card
- Start the instrument as usually
- Start your computer with the GRIMM Software
- Dial up the phone number of the line you connected to the dust monitor via the GRIMM Software
- The instrument will identify the modem and establish the proper communication network (like when sending a email). Than select the operational mode (mostly "on-line") and use the software as usually.
- The instrument will send the results via telephone to your computer and the results will appear and be seen on the display in the pre-selected mode.
- The modem card can also buffer up to 500 data, so there is no need to monitor on line at all times. You may choose a time interval for data transfer via your software.
- Emails can be send to the dust monitor as well, see operating procedure in the key function section. The massage will be displayed when desired.

10 Instrument maintenance and cleaning

It is recommended to work off one time annually following points!

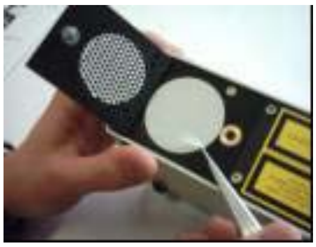
10.1 Filter-Holder Assembly

To make sure that crossover contamination does not occur, the filter holder must be thoroughly cleaned each time a filter is changed. Use a cotton swab or a lint-free cloth to thoroughly clean the entire filter holding assembly and take care that excess debris/lint does not fall into the optical chamber.



Never use any solvents or industrial cleansers to clean the filter housing or stainless steel sampling path.

10.2 Filter changes

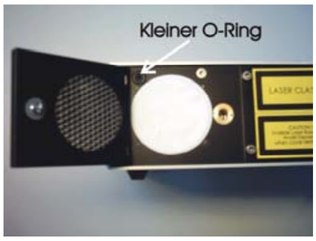


1.) The filter assembly is the square black panel located on the back panel of the instrument. Turn the instrument off. Loosen the screw located on the filter assembly. The screw is attached to the door of the filter assembly in a manner that prevents complete removed of the screw. Once the screw has been completely loosened, turn the unit until the front of the instrument faces upwards. Failure to turn the instrument during the removal of the filter could cause all or part of the collect matter to fall off the filter. Carefully open the filter assembly door. The collected matter is located on the filter between the instrument and the filter.

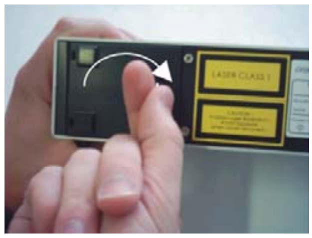
2.) Using pre-cleaned tweezers, carefully remove the filter. Use a lint free cloth to carefully remove any visible material from the filter housing. Great care should be taken to make sure none of the residue falls back into the sampling tube.

3.) Clean the inner stainless steel tube as outlined in and the filter housing as outlined in 10.1 Filter-Holder Assembly

4.) Take a new filter with tweezers and place it in the filter assembly, making sure the large "O" ring is completely covered. The filter should not touch the small "O" Ring. If you want to establish a gravimetric, C-Factor follow the instructions of section 5.5 Gravimetric Factor.



5.) Close the cover carefully so that the filter remains in the correct position. Tighten the screw so that the door to the assembly is closed securely. Do not over tighten the screw.



The operating efficiency of the battery and the life span of the pump are greatly effected by the cleanliness of the filter. It is essential that the filter is changed on a regular basis.

The unit regularly monitors the theoretical total weight collected on the filter. A maximum limit for particulate depends on the kind of aerosol collected. Once this maximum amount has been collected on the filter, the unit will prompt the user to change the filter via the LCD screen. Once this message is displayed, the filter should be changed as soon as possible. In general the maximum amount is approximately 20 mg.

10.3 Cleaning of the measuring chamber

1. Switch off the unit.
2. Open the filter chamber and remove the gravimetric filter as outlined in 10.2 Filter changes
3. With the door open, visually inspect the sample inlet. There should be no visible debris or obstruction. If there is, a gentle stream of clean oil-free air may be used to remove the obstruction. Never use any compressed air above three 3 bars. After cleaning the sample pathway, clean the filter assembly as described in 10.1 Filter-Holder Assembly

CAUTION:

- Never blow air into the sample inlet with the filter assembly door in the closed position. Doing so will damage the internal lenses and may also compromise the sampling system.
- Air must never be blown into the unit from the opening at the filter housing. Clean air must only to be blown into the system in the direction of the normal sampling pathway.
- The sampling path is to be cleaned by first opening the filter assembly and removing the old filter.

10.4 Cleaning the casing

The dust monitor has a metal casing which is a protection against mechanical forces and against influence from EMV. The keypad and the display must also be protected from mechanical stress. To clean the casing only use a dry, clean cloth. Never spill liquid over the instrument.

10.5 Internal rinsing air filter

To protect the measuring chamber and the laser optics from contamination, clean, dry air is feed into the measuring cell. Also to provide clean air for the self test during start up. This clean air is produced by a special rinsing air fine dust filter. This filter has a operating time of several years and should only be changed by trained service personnel. If the system message

Check nozzle **and air inlet !**

is displayed several times, even after cleaning the sample flow path, than an internal failure of the rinsing air supply has occurred. In such a case please contact GRIMM Aerosol Technik or an authorized dealer.

10.6 Change main dust filter

Once a year, the main dust filter (BQ filter), on which all the sampled dust is collect must be exchanged. This will be done during the yearly maintenance, which will be made by GRIMM Aerosol Technik or an authorized dealer.

11 Accessories

To the basic instrument is a wide range of application depending accessories available. The following parts are the most common items requested:

11.1 Sampling probes

Proper sampling is the first step to a successful measurement. Therefore several different methods for IN-DOOR are offered. In general are larger particles above 15 µm diameter not entering easily into any sample pipe, due to their larger mass.
Therefore it should at all times samples only from the top!

Naturally is it possible to sample completely without any of our sampling pipes, but it is possible that larger particles may fall into the "AIR INLET" and they may stick inside the sample tube or the optic.

<u>Depending on the application GRIMM offers therefore different sample methods:</u>

- **Radial symmetric sampler (Model 1.111)**
  This a 7 cm long stainless steel pipe with a random sampling head on top. The air speed (up to 2,5 m/s different direction) entering into a Grimm 1.100 dust monitor at this slit is equal to the inhalation air velocity of a human being (Johannesburg model). This is the most common application for all kind of human studies.

- **Isokinetic sampler (Model 1.151)**
  A representative sample at air speed different to human inhalation requires an isokinetic-sampling head. To sample in this condition it is necessary to know the appropriate wind speed. This unit is suitable for speeds between 0.5 and 10 m/sec since it comes with four different sampling heads. Select the head closest to the wind speed measured [in m/s-value] then sample against the wind direction. In any case you need to sample rectangular to the air flow to make sure all particles enter the sampler opening.

Sampler Colour	Wind speed
-red nozzle:	up to 0,5 m/s
-gold nozzle:	up to 1,0 m/s
-green nozzle:	up to 2,0 m/s
-blue nozzle:	up to 4,0 m/s

PS: Make sure measure "IN COUNTS ONLY"

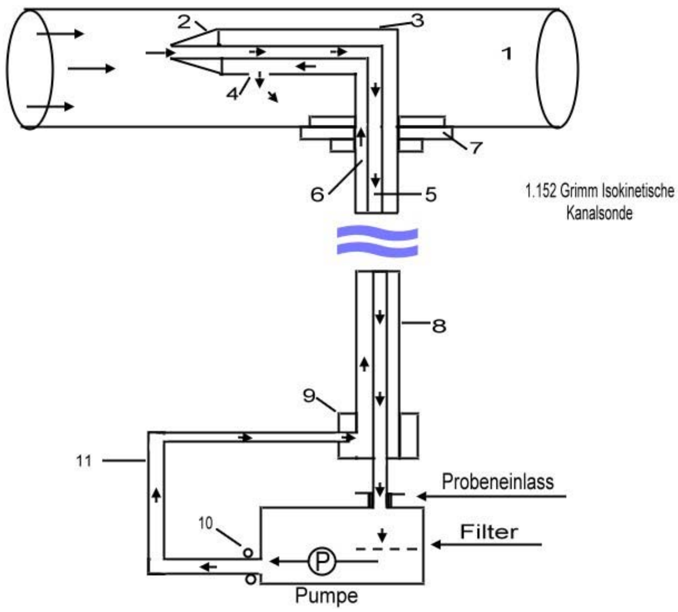
Remember that the sampling inlet must always face the wind. Keep the plastic tube as short as possible and avoid bending it. If possible, have the dust monitor directly below the sampling port. In this way, you may not lose as many of the larger particles due to particle losses through sedimentation.

- **Isokinetic channel probe for 2,5 up to 25 m/s (Model 1.152)**

This sampler is specifically designed for dust-measurements in ventilation canals, at higher wind speeds, pressure, etc. The sampler not only brings the sample air to the monitor, it also collects the exhaust air and conducts it back via the same pipe into the duct. This neutralizes the pressure differential since the built-in sheet-air protection system in the dust monitor is very sensitive to pressure changes and cannot be used above/below 100 mbar differential pressure. If the sampling port to with the sampler will be entered in the air duct, it needs an opening of 35mm in diameter by which to enter. Orienting and scanning through the duct is made easy by adding a scale at the sampling stainless steel pipe (in accordance to VDI 2066).

The 1.152 probe consists of following components:

- (**2**) Four (4)different nozzles for wind speeds of 2-4, 4-8, 8-16 and 16-25 m/sec. The nozzle with the smallest jet opening tube is used for the highest speed.
- (**3**) The 90° probe-sampler. In front is the nozzle needed, at the end this the extension pipe in stainless steel 250 mm long.
- (**7**) Air duct-mounting support
- 0.5 m tubes 3.6 mm for the return air
- (**11**) 1m tubes 3.4 mm for the sample air
- (**9**) Hose-connection-piece (1.119)
- (**10**)Connection for the air return tubes
- Wrench 8*10 for the nozzle-change and to fix the connection for the air return tubes on the back panel (thread hole under the black cover)



Principle of the isokinetic channel probe

11.2 Opto-Acoustic Alarm-Unit (Model 1.147)

This alarm unit is directly connected to the RS-232 port of the dust monitor. Each time the alarm value entered is exceeded, unit gives a signal both visually and audibly. Over five (5) DIL switches and a potentiometer located in the base of the alarm unit give 24 different signal levels to chose from.

Technical Data:

Weight:	0.42 kg
Dimensions:	Size =93 mm, H=115 mm
Plugs:	9 pin D-plug
Power supply:	12V,0.15A
Lighting energy:	1.0 JS
Lighting frequency:	1 Hz
Signal tone:	24 different tones
Volume:	Maximum 110 dB(As)

11.3 Zero-Air-Filter (Model 1.148)

Using this filter, the dust monitor can be checked for possible leaks in the air inlet. Simply connect the filter at the monitor’s air-inlet and the concentration measurements go back to zero. The LCD panel will display this in one (1) minute. The increase of the pump current on and over 60 percent and a possible error message is normal.

11.4 Sensors for temperature and humidity (Model 1.153)

Because of low power consumption this sensor is ideal for the battery-driven dust measuring operations.

Technical data:

Dimensions:	Diameter =15 mm
Length	130 mm
Cables:	ca. 2m
Plugs:	6- pin
Power supply:	10V ±5%, <5 MA
Temp range:	0.3 to +80 °CS
Dissolution:	0.1 ks
Precision:	type. 0.3 ks
Humidity range:	0.1 to 100% rH
Dissolution:	0.1%
Precision:	type. 1%

11.5 Sensors for ambient temperature, humidity and velocity (Model 1.154)

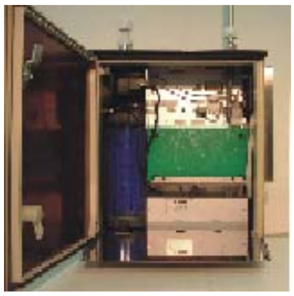
<u>Technical Data:</u>

Dimensions:	∅ = 15 mm
Length	200 mm
Cables:	approx.3m
Plugs:	6- pin
Power supply:	10V ±5%, <5 MA
Temp range:	0.3 to +80 °CS
Dissolution:	0.1 ks
Precision:	type. 0.3 ks
Humidity range:	0.1 to 100% rH
Air velocity	0,3…20 m/s
Dissolution:	0.1%
Precision:	type. 1%

11.6 PAH- Sensor (Model 130)

The PAH sensor qualitatively measures the concentration of particle-bound polycyclic aromatic hydrocarbons (PAH) and is installed in the dust monitors “Model 107-GF” instead of the extern filter. The data transmission in the weather housing accurse over a so called “One-Wire”-Interface. Using the “107-GF” as a portable unit the analog plug can be used for data transmitting. Usually a splitter for the power supply of a GRIMM- dust monitor model 107 is used instead of a power supply.

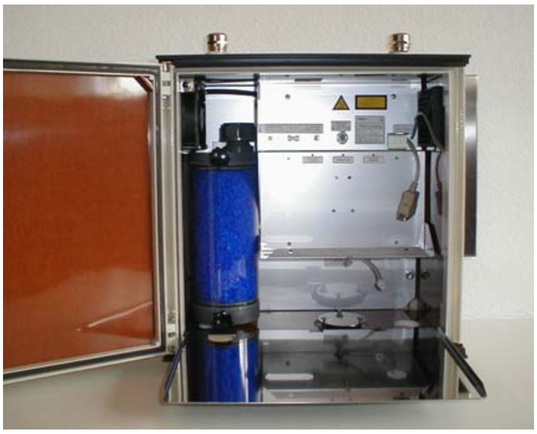
<u>Technical Data:</u>



	240 x 75 x 75 mm
weight	1,8 kg
Plug:	9-pole SUB-D
Power supply:	10 - 28 VDC; < 1,2 A (max.)
Interface:	RS-232, 1-wire, Analog-Out; I2C
Measuring rang:	0 - 5000 fA max. (adjustable)
Precision:	< 1 ng/m^3 (depend on Aerosol)
Data rate:	<= 6 s - 2h (adjustable)
Status display:	LED’s
Dissolution:	>= 12 Bit

Für das Modell 130 ist ein separates ausführliches Handbuch verfügbar.

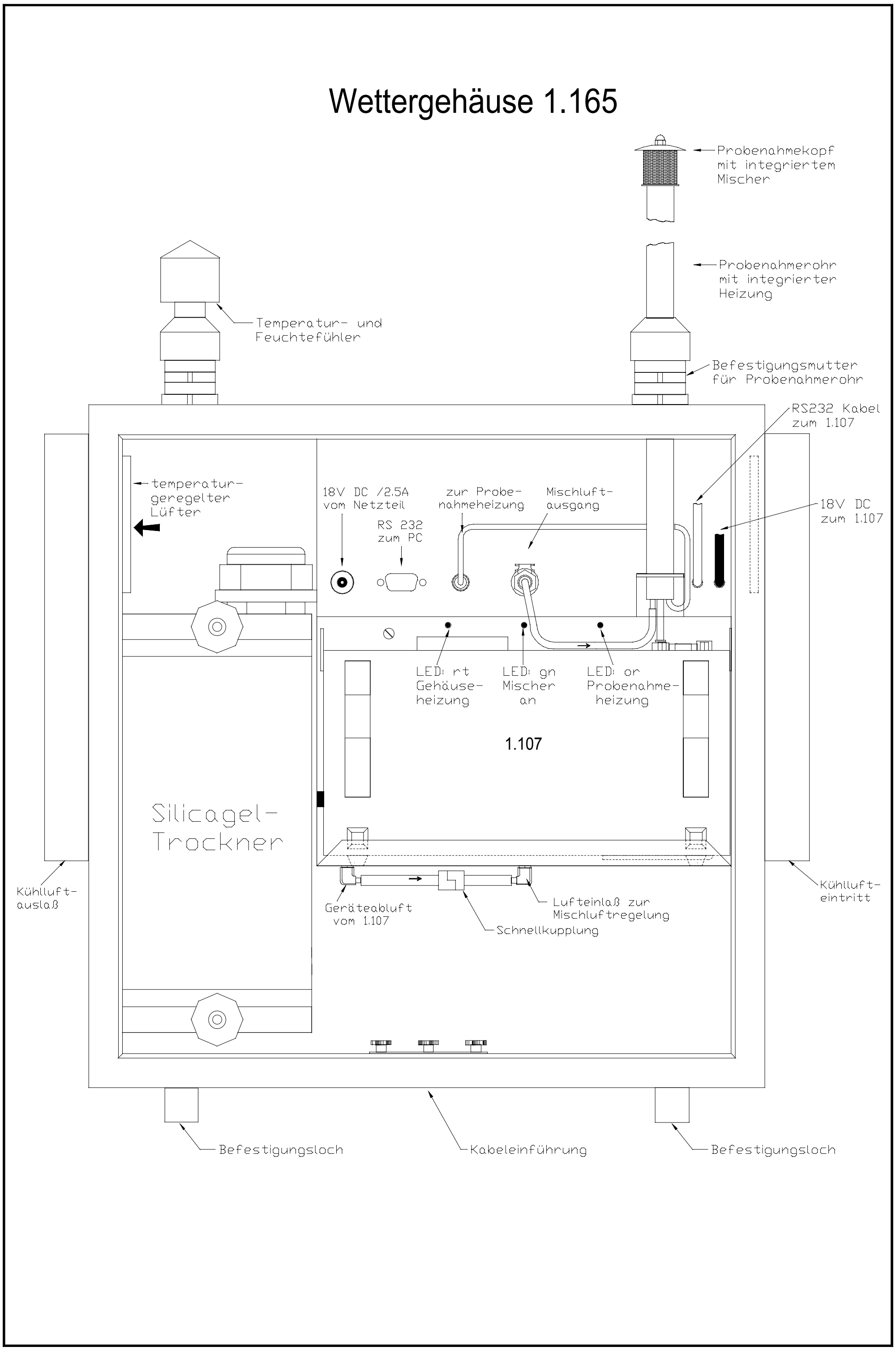
11.7 Weather housing (Model 165FG)



For outdoor measurement the #107 dust monitor has to be protected against climate changes and harsh conditions. The dust monitoring system ENVIRONcheck of Grimm Aerosol Technik works in such a way that the portable 107 monitor will fit perfectly into the outdoor housing and all connections and the logic will work together. The housing itself is absolutely water tight, so rain and other spill can not penetrate the monitoring system. Therefore this unit is ideal for a stand-alone installation as road-side monitor, etc... Since is maintains his internal temperature closed to 22 °C at all times.
High humidity can cause an error in the dust monitor readings because water molecules and nuclear condensation with can misread as solid particles. In such a case the measurement requires a specialty humidifier and this has been done here by using a special dehumidification system build into the weather housing which does not heat or change the sample. Control of the humidity in the model ENVIRONcheck Model #107 is the done with the humidity sensor because the air volume is constant 1.2 liter per minutes and the delusion system which clicks in is in a ration of 1:1 so consequently the counter in the system is counting half and automatically the actor will be multiplied because of the dilution factor 1:1. The air then goes into the mixing chamber in the air inlet pipe and to the dust monitor.

For the Model 165FG there is a separate detailed user manual available!

Housing with dryer and mixing electronic (Model 165) and sample head with integrated mixing chamber (Model 1.170M)



12 Warranty

GRIMM Aerosol Technik, hereinafter referred to as GRIMM, warrants the equipment purchased hereunder to be free from defect in materials and workmanship under normal use and service, when used for the purpose for which it is designed, for a period of (1) one year from the date of shipment. GRIMM further warrants that the equipment will perform in accordance with the technical specific GRIMM owns accompanying the formal equipment offer.

GRIMM will repair or replace any such defective items that may fail within the stated warranty period, PROVIDED:

That any claim of defect under this warranty is made within thirty (30) days after discovery thereof and that inspection by GRIMM , if required, indicates the validity of such claim to GRIMM 's faction; and

- That the defect is not the result of damage incurred in shipment to or from our factory; and
- That the equipment has not been altered in any way whether as to design or use, whether by replacement parts not supplied or approved by GRIMM , or otherwise; and
- That any equipment or accessories furnished but not manufactured by GRIMM , or not of GRIMM design, shall be subject only to such adjustments as GRIMM may obtain from the supplier thereof.

GRIMM 's obligation on under this warranty is limited to the repair or replacement of defective parts with the exception noted above. If the equipment includes a scattering chamber, GRIMM 's warranty does not extend to contamination on of the scattering chamber by foreign material.

At GRIMM 's option, any defective equipment that fails within the warranty period shall be returned to Grimm's factory for inspection, properly packed with shipping charges prepaid. No equipment shall be returned to GRIMM without prior issuance of a return authorization on by GRIMM .

No warranties, express or implied, other than those specifically set forth herein shall be applicable to any equipment manufactured or furnished by GRIMM and the foregoing warranty shall constitute the Buyer's sole right and remedy. In no event does GRIMM assume any liability for consequential damages, or for loss, damage or expense directly or indirectly arising from the use of GRIMM products, or any inability to use them either separately or in combination on with other equipment or materials or from any other cause.

13 Shipment

All equipment described in this document is shipped inside a container or a special packing.

Check shipment for completeness and visual damages.

Any damage found has to be reported immediately to the supplier.

In this case the equipment must **N O T** be taken into operation for safety reasons.

To protect the equipment against potential damage it is recommended to keep the original packing material.

Any transport lock mechanism must be activated prior to shipment in case that the instrument is equipped with such a feature.

If the equipment has been exposed to low temperatures during transportation, one must allow the equipment to stabilize to room temperature, before taking it into operation. Otherwise malfunction or damage could result.

14 Repair

GRIMMS policy is to serve customer problems which are related to instrument malfunction as quick as possible.
Please contact GRIMM or it's representative in case of a malfunction.
Send an Email prior to shipment to

Service@grimm-aerosol.com

and provide the following information :

- Instrument- Model- Number
- Serial- Number and date of manufacture (see identification plate)
- Date of order and PO- number (except it's warranty)
- Invoice Address
- Shipping Address

The customer must guarantee that any equipment shipped to GRIMM is free of any hazardous contamination !

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